HealthConnect Rwanda: Healthcare Analytics & Predictive Modeling Project

> **Track:** Data Science
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> **Program:** AnalystLab Africa Internship
> **Project Phase:** Week 6 - HealthConnect Rwanda: Healthcare Analytics & Predictive Modeling Project

Project Overview

This notebook serves as the comprehensive data science environment for the HealthConnect Rwanda Healthcare Analytics and Artificial Intelligence Transformation Project (Path 2: Healthcare Predictive Analytics and Population Health Intelligence).

HealthConnect Rwanda manages a vast ecosystem of clinical, operational, financial, and administrative data across 30 logical tables assembled from CSV files and multi-sheet Excel workbooks. This project aims to extract actionable insights, build predictive models for patient admissions and hospital revenues, perform advanced data cleaning, and formulate strategic recommendations to optimize healthcare delivery and resource allocation.

Notebook Structure

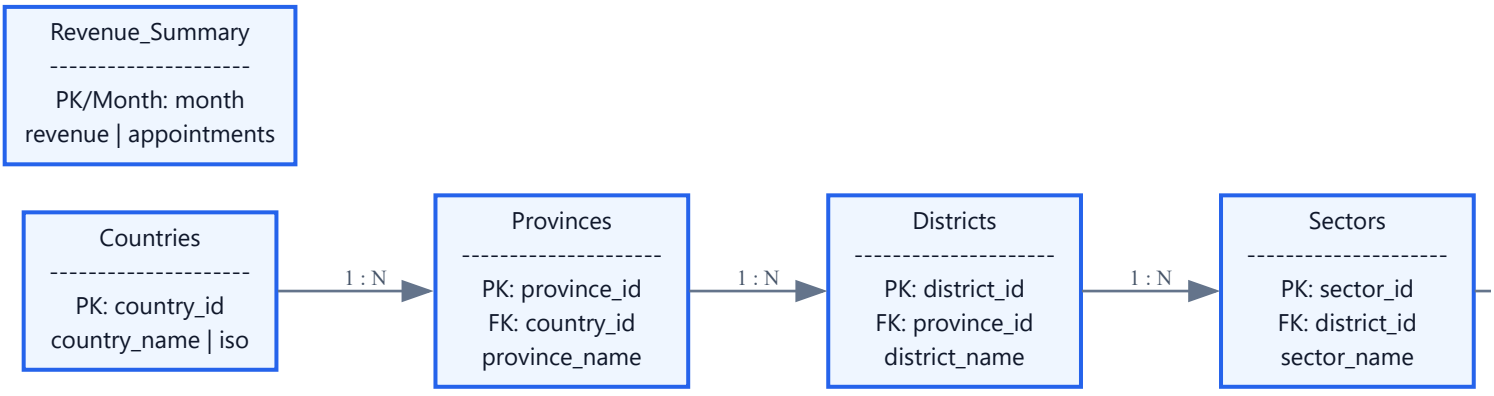
- **Part 1:** Data Acquisition, Structural Understanding & Entity Relationship Modeling (ERD)
- **Part 2:** Data Cleaning & Preprocessing
- **Part 3:** Data Validation & Quality Assessment
- **Part 4:** Exploratory Data Analysis (EDA)
- **Part 5:** Feature Engineering
- **Part 6:** Statistical Analysis & Hypothesis Testing
- **Part 7:** Predictive Analytics, Forecasting & Clustering
- **Part 8:** Business Insights & Strategic Recommendations

Part 1: Data Acquisition & Structural Understanding

In this first section, we ingest all logical datasets provided by HealthConnect Rwanda, including CSV files and multi-sheet Excel workbooks. We inspect their dimensions, data types, and structural integrity. Following ingestion, we utilize Graphviz to construct and visualize an Entity Relationship Diagram (ERD), mapping the primary and foreign key relationships across tables to establish a clear structural foundation for subsequent analytical modeling.

```
Loaded 30 logical tables from 37 source files/sheets.
```

	Table_Name	Source_Count	Source_Files	Rows_Across_Sources	Exact_Cr
0	Administrative_Staff	1	HC_Administrative_Staff.csv	80	
1	Admissions	1	HC_Clinical_Transactions.xlsx::Admissions	5000	
2	Appointments	1	HC_Clinical_Transactions.xlsx::Appointments	5000	
3	Beds	2	HC_Beds.csv | HC_Facility_Master_Data.xlsx::Beds	4000	
4	Billing	1	HC_Billing.csv	60000	
5	Countries	1	HCCountries.csv	1	
6	Departments	2	HC_Departments.csv | HC_Facility_Master_Data.x...	400	
7	Diagnoses	2	HC_Clinical_Transactions.xlsx::Diagnoses | HC_...	65000	
8	Districts	1	HC_Districts.csv	30	
9	Doctor	1	HC_Doctor.csv	350	
10	Insurance_Claims	1	Insurance_Claims.csv	42036	
11	Laboratory_Scientist	1	Laboratory_Scientist.csv	40	
12	Laboratory_Tests	1	HC_Clinical_Transactions.xlsx::Laboratory_Tests	5000	
13	Medical_Equipment	2	HC_Facility_Master_Data.xlsx::Medical_Equipmen...	400	
14	Medical_Specialties	2	HC_Facility_Master_Data.xlsx::Medical_Specialt...	400	
15	Nurse	1	HC_Nurse.csv	700	
16	Ownership_Types	1	HC_Ownership_Types.csv	4	
17	Patient_Feedback	1	HC_Patient_Feedback.csv	21064	
18	Patients	1	HC_Patients.csv	20000	
19	Payments	1	HC_Payments.csv	60000	
20	Pharmacist	1	Pharmacist.csv	50	
21	Prescriptions	1	HC_Clinical_Transactions.xlsx::Prescriptions	5000	
22	Provinces	1	HC_Provinces.csv	5	
23	Radiographer	1	HC_Radiographer.csv	30	
24	Referrals	1	HC_Clinical_Transactions.xlsx::Referrals	4708	
25	Revenue_Summary	1	HC_Revenue_Summary.csv	12	
26	Rooms	2	HC_Facility_Master_Data.xlsx::Rooms | HC_Rooms...	800	
27	Sectors	1	HC_Sectors.csv	120	
28	Services	2	HC_Facility_Master_Data.xlsx::Services | HC_Se...	800	
29	Villages	1	HC_Villages.csv	480	


Revenue_Summary
PK/Month: month
revenue | appointments
Countries
PK: country_id
country_name | iso
1 : N
Provinces
PK: province_id
FK: country_id
province_name
1 : N
Districts
PK: district_id
FK: province_id
district_name
1 : N
Sectors
PK: sector_id
FK: district_id
sector_name

Part 2: Data Cleaning & Preprocessing

2.1 - Data Integration at the Appointment Grain

Integration principle

The source tables describe different entities and have different levels of detail. A direct join of every table would multiply appointment rows when an appointment has several diagnoses, tests, prescriptions, bills, payments, or feedback records. The analytical master table will therefore use **one row per appointment**. One-to-many clinical and financial tables will be summarized to appointment level before they are joined.

The original relational tables remain available in `datasets` for traceability. Cleaning is applied to the integrated appointment-level master table used in the downstream analysis. The master table excludes direct patient identifiers

such as names, phone numbers, and emergency contacts. Source-level duplicate copies are combined during ingestion; exact repeated rows are removed, while distinct records are retained for quality review.

```
Master analytical table: 5,000 appointments and 63 columns.
```

	AppointmentID	PatientID	DoctorID	FacilityID	DepartmentID	Date	VisitType	Gender	DateOfBirth	Age
0	APT000001	PAT18724	DO0017	HF014	DEP124	2026-01-07	Follow-up	Male	2014-01-01	12
1	APT000002	PAT10741	DO0039	HF008	DEP191	2026-01-14	Emergency	Male	1948-01-01	78
2	APT000003	PAT09933	DO0339	HF012	DEP035	2026-04-15	Emergency	Male	1992-01-01	34
3	APT000004	PAT11998	DO0123	HF011	DEP171	2026-08-14	Follow-up	Female	1993-01-01	33
4	APT000005	PAT13529	DO0123	HF002	DEP009	2026-05-27	Emergency	Male	1939-01-01	87

5 rows × 63 columns

2.2 - Pre-cleaning Quality Profiling

This profiling step measures source quality before cleaning so preprocessing decisions are based on observed missingness, duplication, validity, and relationship coverage. Formal post-clean validation is reported in Part 3. It covers completeness, exact duplicates, primary-key uniqueness, invalid values, data types, date parsing, and foreign-key integrity for relationships whose parent tables are present.

The assignment asks for facility and department ID validation. Department IDs can be checked against the department table. The supplied folder does not contain a facility master table, so facility IDs will be reported as **not independently verifiable** rather than treated as valid or invalid by assumption.

	Table_Name	Rows	Columns	Missing_Cells	Missing_Cell_Rate_Pct	Rows_With_Missing	Exact_Duplicate_
0	Administrative_Staff	80	10	0	0.00	0	
1	Admissions	5000	3	0	0.00	0	
2	Appointments	5000	7	0	0.00	0	
3	Beds	2000	3	0	0.00	0	
4	Billing	60000	4	0	0.00	0	
5	Countries	1	6	0	0.00	0	
6	Departments	200	3	0	0.00	0	
7	Diagnoses	60000	4	0	0.00	0	
8	Districts	30	3	0	0.00	0	
9	Doctor	350	10	0	0.00	0	
10	Insurance_Claims	42036	5	0	0.00	0	
11	Laboratory_Scientist	40	10	0	0.00	0	
12	Laboratory_Tests	5000	4	0	0.00	0	
13	Medical_Equipment	200	4	0	0.00	0	
14	Medical_Specialties	200	3	0	0.00	0	
15	Nurse	700	10	0	0.00	0	
16	Ownership_Types	4	2	0	0.00	0	
17	Patient_Feedback	21064	4	0	0.00	0	
18	Patients	20000	19	5071	1.33	5071	
19	Payments	60000	4	0	0.00	0	
20	Pharmacist	50	10	0	0.00	0	
21	Prescriptions	5000	4	0	0.00	0	
22	Provinces	5	3	0	0.00	0	
23	Radiographer	30	10	0	0.00	0	
24	Referrals	4708	3	0	0.00	0	
25	Revenue_Summary	12	3	0	0.00	0	
26	Rooms	400	4	0	0.00	0	
27	Sectors	120	3	0	0.00	0	
28	Services	400	3	0	0.00	0	
29	Villages	480	3	0	0.00	0	

	Table_Name	Column_Name	Data_Type	Missing_Count	Missing_Rate_Pct	Distinct_Values	Blank_String_Coun
98	Patients	ChronicCondition	str	5071	25.36	3	

	Table_Name	Key_Column	Status	Null_Keys	Duplicate_Key_Rows
0	Countries	CountryID	Evaluated	0	0
1	Provinces	ProvinceID	Evaluated	0	0
2	Districts	DistrictID	Evaluated	0	0
3	Sectors	SectorID	Evaluated	0	0
4	Villages	VillageID	Evaluated	0	0
5	Ownership_Types	OwnershipID	Evaluated	0	0
6	Departments	DepartmentID	Evaluated	0	0
7	Medical_Specialties	SpecialtyID	Evaluated	0	0
8	Services	ServiceID	Evaluated	0	0
9	Medical_Equipment	EquipmentID	Evaluated	0	0
10	Rooms	RoomID	Evaluated	0	0
11	Beds	BedID	Evaluated	0	0
12	Doctor	StaffID	Evaluated	0	0
13	Nurse	StaffID	Evaluated	0	0
14	Pharmacist	StaffID	Evaluated	0	0
15	Laboratory_Scientist	StaffID	Evaluated	0	0
16	Radiographer	StaffID	Evaluated	0	0
17	Administrative_Staff	StaffID	Evaluated	0	0
18	Patients	PatientID	Evaluated	0	0
19	Appointments	AppointmentID	Evaluated	0	0
20	Diagnoses	DiagnosisID	Evaluated	0	0
21	Laboratory_Tests	LabID	Evaluated	0	0
22	Prescriptions	PrescriptionID	Evaluated	0	0
23	Admissions	AdmissionID	Evaluated	0	0
24	Referrals	ReferralID	Evaluated	0	0
25	Billing	BillingID	Evaluated	0	0
26	Insurance_Claims	ClaimID	Evaluated	0	0
27	Payments	PaymentID	Evaluated	0	0
28	Patient_Feedback	FeedbackID	Evaluated	0	0

	Child_Table	Foreign_Key	Parent_Table	Parent_Key	Status	Non_Null_Keys	Orphan_Keys
0	Provinces	CountryID	Countries	CountryID	Evaluated	5.0	0.0
1	Districts	ProvinceID	Provinces	ProvinceID	Evaluated	30.0	0.0
2	Sectors	DistrictID	Districts	DistrictID	Evaluated	120.0	0.0
3	Villages	SectorID	Sectors	SectorID	Evaluated	480.0	0.0
4	Patients	VillageID	Villages	VillageID	Evaluated	20000.0	0.0
5	Departments	FacilityID	Facilities master not supplied	FacilityID	Not evaluable	NaN	NaN
6	Medical_Specialties	DepartmentID	Departments	DepartmentID	Evaluated	200.0	0.0
7	Services	DepartmentID	Departments	DepartmentID	Evaluated	400.0	0.0
8	Medical_Equipment	DepartmentID	Departments	DepartmentID	Evaluated	200.0	0.0
9	Rooms	DepartmentID	Departments	DepartmentID	Evaluated	400.0	0.0
10	Beds	RoomID	Rooms	RoomID	Evaluated	2000.0	0.0
11	Appointments	PatientID	Patients	PatientID	Evaluated	5000.0	0.0
12	Appointments	DoctorID	Doctor	StaffID	Evaluated	5000.0	0.0
13	Appointments	DepartmentID	Departments	DepartmentID	Evaluated	5000.0	0.0
14	Diagnoses	AppointmentID	Appointments	AppointmentID	Evaluated	60000.0	55000.0
15	Laboratory_Tests	AppointmentID	Appointments	AppointmentID	Evaluated	5000.0	2728.0
16	Prescriptions	AppointmentID	Appointments	AppointmentID	Evaluated	5000.0	997.0
17	Admissions	AppointmentID	Appointments	AppointmentID	Evaluated	5000.0	4414.0
18	Referrals	AppointmentID	Appointments	AppointmentID	Evaluated	4708.0	4293.0
19	Billing	AppointmentID	Appointments	AppointmentID	Evaluated	60000.0	55000.0
20	Payments	BillingID	Billing	BillingID	Evaluated	60000.0	0.0
21	Insurance_Claims	BillingID	Billing	BillingID	Evaluated	42036.0	0.0
22	Patient_Feedback	AppointmentID	Appointments	AppointmentID	Evaluated	21064.0	19308.0

	Table_Name	Column_Name	Check	Flagged_Rows
0	Patients	Age	Age must be between 0 and 120 years	0
1	Billing	Amount	Billing amount must be non-negative	0
2	Payments	AmountPaid	Payment amount must be non-negative	0
3	Insurance_Claims	ClaimAmount	Claim amount must be non-negative	0
4	Admissions	LengthOfStayDays	Length of stay must be non-negative	0
5	Patient_Feedback	Rating	Feedback rating must be between 1 and 5	0
6	Doctor	YearsExperience	Experience must be non-negative	0
7	Appointments	Date	Non-empty values that cannot be parsed as dates	0
8	Patients	DateOfBirth	Non-empty values that cannot be parsed as dates	0
9	Revenue_Summary	Month	Non-empty values that cannot be parsed as dates	0

2.3 - Master DataFrame Cleaning and Preprocessing

The cleaning target is the integrated, appointment-level `master_df`, which is the analytical dataset used in the next parts of this notebook. The source tables are retained unchanged in `datasets` for traceability and for later domain-specific analyses; they are not separately exported as cleaned datasets.

Missingness is interpreted by field meaning. Absence of a related event is represented with a zero count or an explicit `No record` label. A missing chronic-condition value is labelled `Not recorded`, because the source does not prove that the patient has no condition. Appointments without feedback receive the `Not rated` satisfaction category; their scores are not fabricated. Required identifiers and dates are validated rather than imputed. The cleaned master should have no null values, while explicit labels preserve what was not observed.

	Rule	Affected_Rows
0	Remove exact duplicate master rows	0
1	Standardize master column names	63
2	Normalize text whitespace and selected categories	Applied to text columns
3	Represent missing chronic-condition values as ...	1263
4	Represent missing child-event counts as zero	Applied to event-count columns
5	Represent absent child-event descriptions expl...	Applied to event-summary columns
6	Categorize absent satisfaction feedback as Not...	3244

	Rows_Before	Rows_After	Columns_After	Null_Cells_After	Duplicate_Appointment_IDs_After	Invalid_Age_Rows
0	5000	5000	63	0	0	0

	appointmentid	patientid	doctorid	facilityid	departmentid	date	visittype	gender	dateofbirth	age	...
0	APT000001	PAT18724	DO0017	HF014	DEP124	2026-01-07	Follow-Up	Male	2014-01-01	12	...
1	APT000002	PAT10741	DO0039	HF008	DEP191	2026-01-14	Emergency	Male	1948-01-01	78	...
2	APT000003	PAT09933	DO0339	HF012	DEP035	2026-04-15	Emergency	Male	1992-01-01	34	...
3	APT000004	PAT11998	DO0123	HF011	DEP171	2026-08-14	Follow-Up	Female	1993-01-01	33	...
4	APT000005	PAT13529	DO0123	HF002	DEP009	2026-05-27	Emergency	Male	1939-01-01	87	...

5 rows × 63 columns

2.4 - Data Quality Report: Scope and Cleaning Evidence

The pre-clean audit profiles the supplied logical tables and validates the available primary-key and foreign-key relationships. The cleaning log applies corrections only to the integrated appointment-level master table. The original source tables are retained unchanged in `datasets` so the source values remain traceable.

The post-clean summary must show zero null cells and unique `appointmentid` values in `master_df_clean`. A zero-null master does not mean every source value was observed: labels such as `Not recorded`, `No record`, and `Not rated` preserve that distinction. Numeric satisfaction ratings remain in the original `Patient_Feedback` table and should be used only for appointments with observed feedback. Facility foreign keys remain not independently verifiable because the supplied project data has no facility master table.

Part 3: Data Validation & Quality Assessment

Part 2 profiled the source tables and cleaned the appointment-level master. This section independently validates the resulting analytical dataset and checks key relationships against source tables. It separates structural failures from findings requiring source-owner review, so inconsistencies are not silently interpreted as valid business facts.

3.1 - Structural, Referential, and Domain Validation

The checks verify the one-row-per-appointment grain, key uniqueness, completeness after cleaning, expected identifier coverage, value domains, and financial non-negativity. Department-to-facility mapping and billing-status/payment inconsistencies are reported for review, not automatically corrected. Events outside the appointment cohort are counted separately because they may belong to a wider extract rather than represent broken keys.

	Validation	Observed	Expected	Result	Interpretation
0	Master row count	5,000	5,000	Pass	The master should retain one row per source ap...
1	AppointmentID uniqueness	5,000 unique IDs	5,000	Pass	One row per appointment is required.
2	Master completeness	0 null cells	0 null cells	Pass	Not recorded labels preserve unavailable sourc...
3	Appointment key-set agreement	5,000 IDs	5,000 source IDs	Pass	Checks whether integration dropped or added ap...
4	Patient foreign keys	0 unmatched appointment rows	0 unmatched rows	Pass	Checked against the corresponding parent table.
5	Doctor foreign keys	0 unmatched appointment rows	0 unmatched rows	Pass	Checked against the corresponding parent table.
6	Department foreign keys	0 unmatched appointment rows	0 unmatched rows	Pass	Checked against the corresponding parent table.
7	Age domain	0 values outside 0-120	0 invalid values	Pass	Age zero is retained as a valid infant age.
8	Non-negative financial totals	0 negative values	0 negative values	Pass	Currency is unspecified; source units are pres...
9	Appointment department/facility consistency	4,744 of 5,000 disagree	0 disagreements	Review	Review source mappings; no values were overwri...
10	Billing status vs payment records	323 Paid bills show a balance; 1,626 Pending b...	Status and transactions reconcile	Review	Use transaction amounts for receipts and balan...

	Table	Source rows	Rows linked to master	Rows outside cohort
0	Diagnoses	60000	5000	55000
1	Laboratory_Tests	5000	2272	2728
2	Prescriptions	5000	4003	997
3	Admissions	5000	586	4414
4	Referrals	4708	415	4293
5	Billing	60000	5000	55000
6	Patient_Feedback	21064	1756	19308

Interpretation & Validation Findings

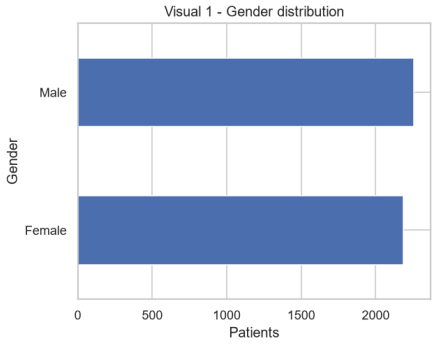
- **Master structure:** There are 5,000 rows for 5,000 unique appointments and no residual null cells after cleaning. Explicit labels such as `Not recorded`, `No record`, and `Not rated` still preserve missing-source meaning.
- **Core relationships:** Patient, doctor, and department identifiers match their available parent tables, and the appointment ID set agrees with the source. The selected appointment grain is intact.
- **Department/facility mapping requires review:** 4,744 of 5,000 appointments (94.9%) have a FacilityID that differs from the FacilityID listed for the same DepartmentID in `Departments`. Both source values remain unchanged. Facility-level attribution should be treated cautiously until the mapping is reconciled.
- **Billing status conflicts with transactions:** Paid bills retain calculated balances, while Pending bills also have linked payments. The transaction amounts are used for receipt and balance calculations; the status labels remain findings for source-owner review.
- **Out-of-cohort events:** The scope table counts child records linked to the 5,000 appointments and those outside the cohort. Out-of-cohort records are excluded from appointment-level analyses, but are not automatically called invalid because they may belong to a wider extract.
- **Validation result:** Structural checks, core keys, age range, and non-negative amounts pass. Department/facility alignment and billing status/payment alignment remain review findings.

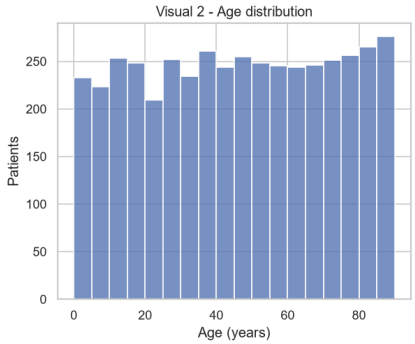
Part 4: Exploratory Data Analysis (EDA)

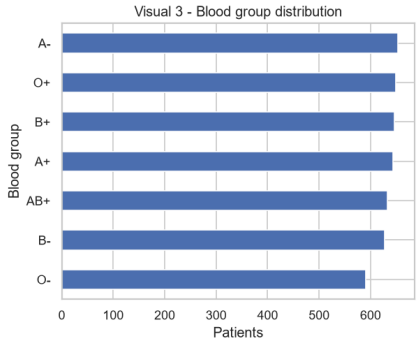
This part explores patient profiles, clinical activity, operational capacity, financial performance, and patient experience using the cleaned appointment-level master table. Detailed event tables are restricted to appointment identifiers present in `master_df_clean` before analysis. Each chart is titled and labeled, and its interpretation highlights both the observed pattern and relevant data limitations.

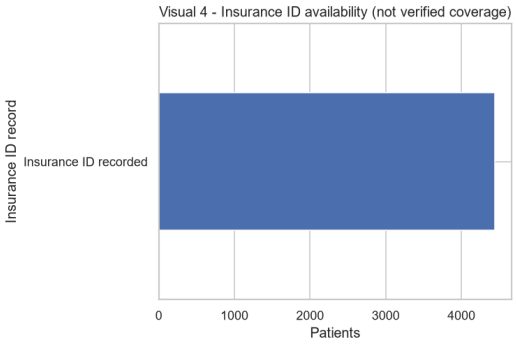
4.1 - Patient Analysis

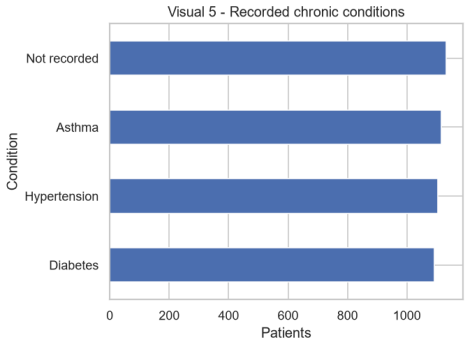
This section describes the demographic and health profile of patients represented in the appointment cohort. Patient growth is measured as first observed appointments by month because the source data do not provide a patient registration date. Geographic results are summarized by administrative area; map boundaries and coordinates are not supplied.

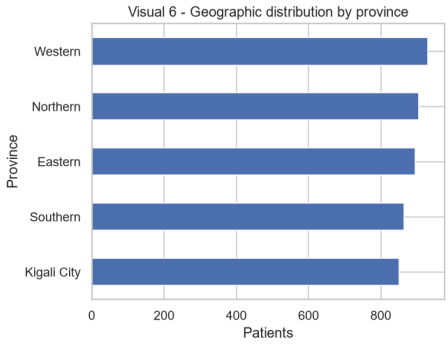


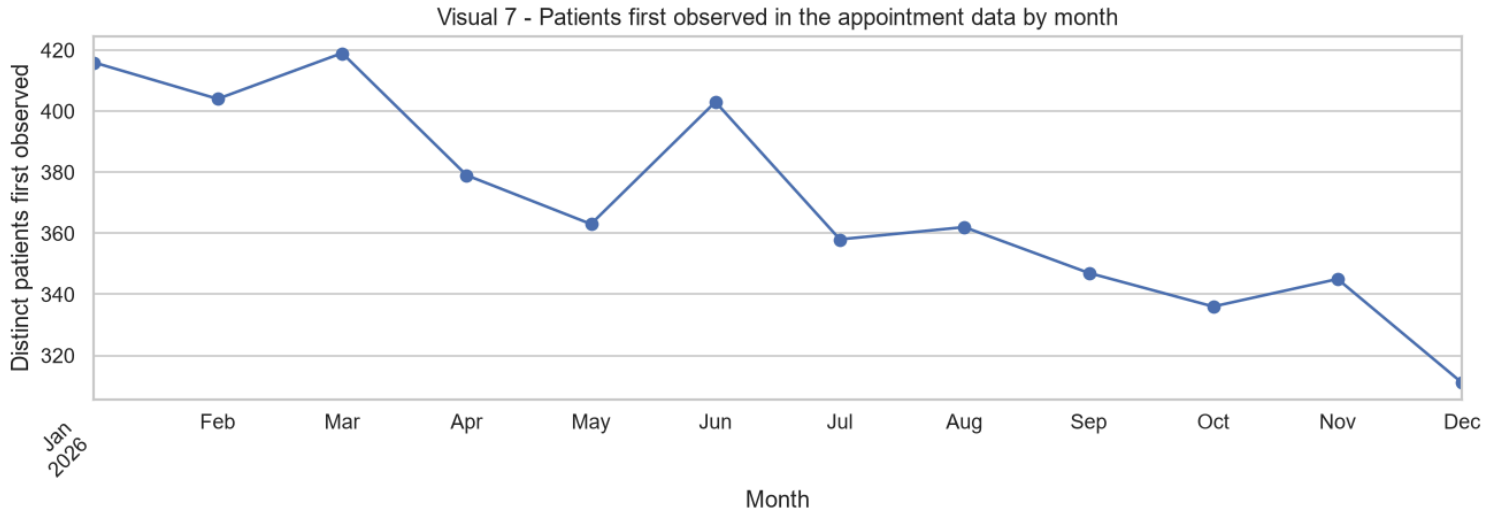













```
Appointment cohort: 5,000 appointments; 4,443 distinct patients.
Age summary (years), one row per patient:
```

	count	mean	std	min	25%	50%	75%	max
age	4443.0	45.517893	26.149465	0.0	23.0	46.0	68.0	90.0

```
Patients with a recorded chronic condition: 3,312 / 4,443 (74.5%); missing condition history is exclude
d from this prevalence estimate.
```

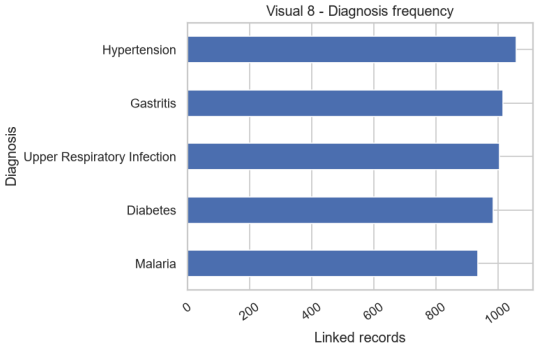
Interpretation & Patient Insights - Visuals 1 to 7

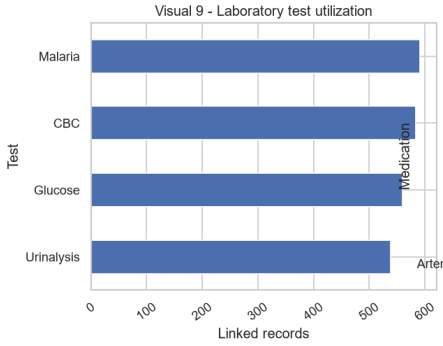
- **Visual 1 (Gender Distribution):** The 4,443 distinct patients are almost evenly divided: 2,256 male (50.8%) and 2,187 female (49.2%). The difference is only 69 patients, so neither gender dominates this appointment cohort.
- **Visual 2 (Age Distribution):** The histogram spans ages 0 to 90 rather than concentrating in one narrow age band. The median is 46 years, the mean is 45.5, and the middle half of patients falls between 23 and 68 years. This indicates that the service cohort includes children and older adults as well as working-age patients; the chart describes patients with appointments, not Rwanda's age structure.
- **Visual 3 (Blood Group Distribution):** Counts are fairly close across the seven recorded blood groups, from 591 for O- to 653 for A-. The 62-patient spread is small relative to the cohort, so there is no single blood group that dominates the recorded patient profile.

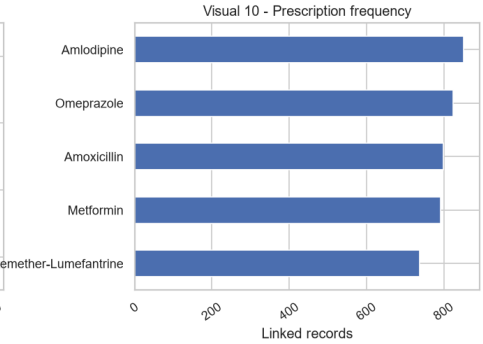
- **Visual 4 (Insurance ID Availability):** All 4,443 distinct patients have an InsuranceID recorded, which is why the chart has one bar. This confirms identifier completeness in the selected cohort only. There is no insurance-status or insurer lookup table here, so the chart cannot establish active coverage, eligibility, or benefit level.
- **Visual 5 (Recorded Chronic Conditions):** Among patients with a recorded condition, counts are similar: Asthma 1,116, Hypertension 1,104, and Diabetes 1,092. However, 1,131 patients (25.5%) have no recorded chronic-condition history. The three condition counts must therefore be read as recorded categories, not confirmed prevalence; missing history may change their true proportions.
- **Visual 6 (Geographic Distribution by Province):** Western has the largest cohort count (930; 20.9%) and Kigali City the smallest (850; 19.1%). The 80-patient gap across the five provinces suggests broadly even representation in these records. Because no provincial population denominators are available, this is not a comparison of healthcare access or population-level rates.
- **Visual 7 (Patients First Observed by Month):** The series peaks in March at 419 first-observed patients and reaches its low in December at 311, a 25.8% difference from the March peak. Counts generally ease after March, with a temporary rebound in June (403). Since there is no patient registration date, this is only the first appointment date observed in the supplied data; it should not be described as actual patient growth or a confirmed seasonal pattern.

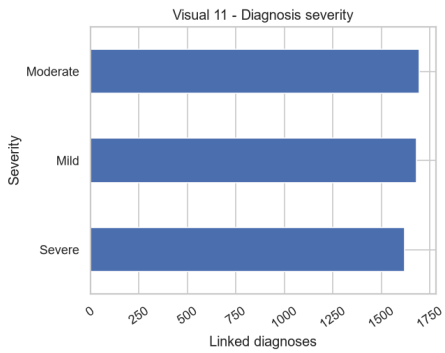
4.2 - Clinical Analysis

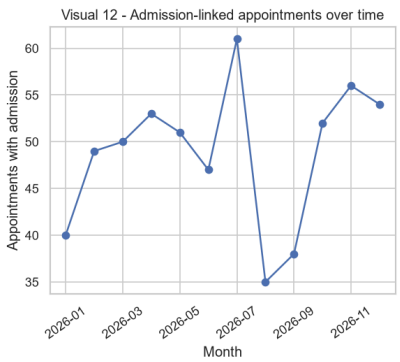
Clinical event tables can contain records outside the appointment cohort and may contain multiple rows per appointment. The analyses below first keep only events linked to the cleaned master appointment identifiers, then count events or appointments according to the chart's stated unit. Severity is descriptive and is not treated as a validated clinical outcome.

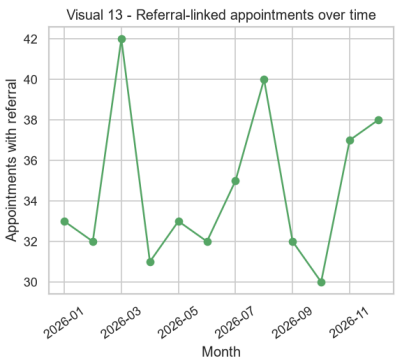













```
Cohort-filtered clinical record counts:
Diagnoses: 5,000 rows
Laboratory_Tests: 2,272 rows
Prescriptions: 4,003 rows
Admissions: 586 rows
Referrals: 415 rows
```

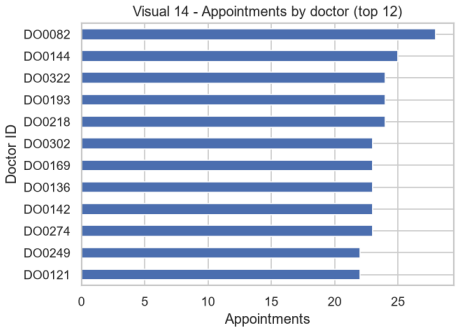
Interpretation & Clinical Insights - Visuals 8 to 13

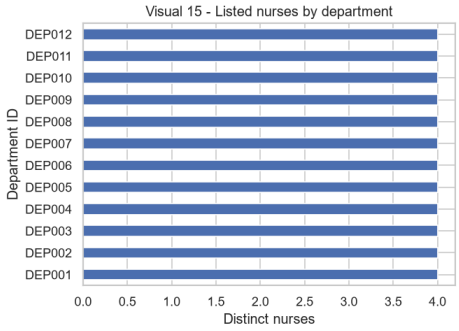
- **Visual 8 (Diagnosis Frequency):** All 5,000 cohort appointments have a linked diagnosis. The five diagnoses occur in a relatively narrow range: Hypertension leads with 1,059 (21.2%), followed by Gastritis 1,017 (20.3%), Upper Respiratory Infection 1,004 (20.1%), Diabetes 985 (19.7%), and Malaria 935 (18.7%). No diagnosis overwhelms the distribution. These frequencies describe the records provided, not disease incidence in the wider population.
- **Visual 9 (Laboratory Test Utilization):** Laboratory records are linked to 2,272 appointments, or 45.4% of the 5,000-appointment cohort. The four test categories are also quite balanced: Malaria 591 (26.0%), CBC 583 (25.7%), Glucose 560 (24.6%), and Urinalysis 538 (23.7%) of linked test records. The bar chart counts test records, not test results, abnormal findings, or clinical necessity.
- **Visual 10 (Prescription Frequency):** Prescriptions appear for 4,003 appointments (80.1%). Amlodipine is the most frequent medication (851; 21.3% of prescription records) and Artemether-Lumefantrine the least frequent among the listed top categories (738; 18.4%). The close spacing indicates no one medication dominates this dataset. Counts do not measure dosage, adherence, treatment success, or appropriateness.
- **Visual 11 (Diagnosis Severity):** Moderate (1,696; 33.9%), Mild (1,682; 33.6%), and Severe (1,622; 32.4%) labels are nearly evenly represented. The small difference between the largest and smallest categories is 74 records. This balance may reflect how the source data were generated or recorded; severity labels have not been clinically validated.
- **Visual 12 (Admission-Linked Appointments by Month):** The chart counts distinct appointments with an admission record. Monthly counts range from 35 in August to 61 in July, with a noticeable July-to-August drop and a rebound to 52 in October and 56 in November. The variation does not form a steady upward or downward trend. This is not a count of admission days or bed occupancy.
- **Visual 13 (Referral-Linked Appointments by Month):** Referral-linked appointments range from 30 in October to 42 in March. The monthly line fluctuates, with a March peak (42) and a second-highest month in August (40); it does not show a sustained trend. The source only records referral linkage, so the chart says nothing about referral completion or destination outcomes.

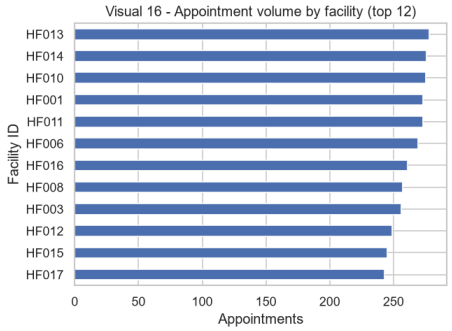
Cohort scope: Clinical tables were filtered to the 5,000 AppointmentIDs in `master_df_clean`; unmatched source records are intentionally excluded from these cohort charts.

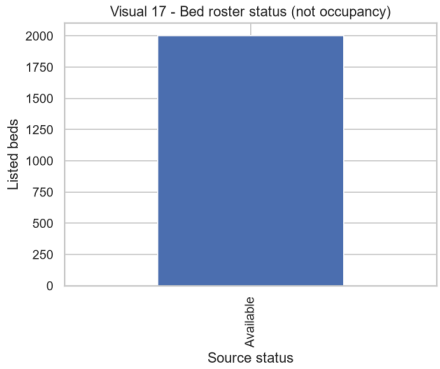
4.3 - Operational Analysis

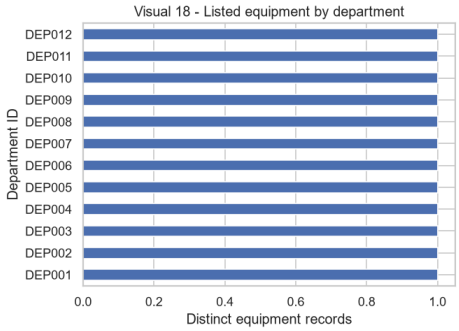
Operational indicators are calculated from appointment volume and staff or equipment rosters. The available files do not assign nurses to individual appointments, provide bed occupancy events, or identify equipment usage; therefore, roster counts and appointment volumes are used as operational context rather than measured workload or utilization rates.

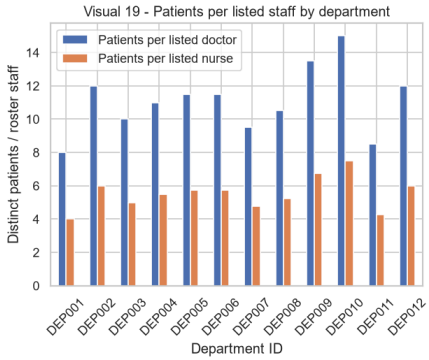













```
Listed beds: 2,000; source status values: {'Available': 2000}
```

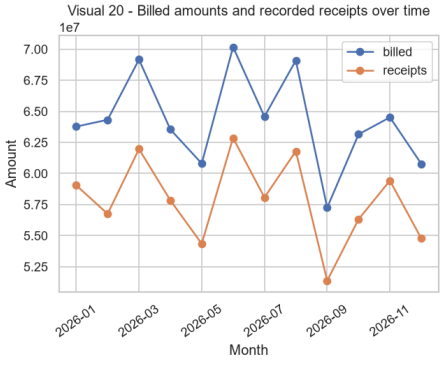
Interpretation & Operational Insights - Visuals 14 to 19

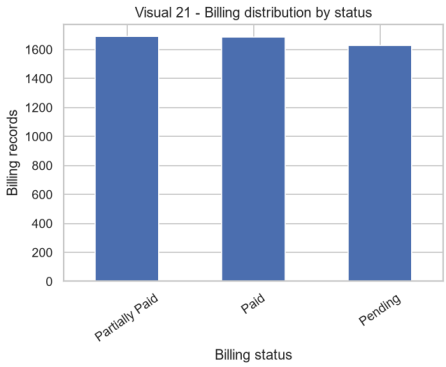
- **Visual 14 (Appointments by Doctor):** The 12 highest-volume doctors each have between 22 and 28 appointments in this cohort; the leading doctor has 28. Across all 350 listed doctors the mean is 14.3 cohort appointments, with a range of 5 to 28. This shows variation in observed appointment counts, but no shift length, working hours, specialty case mix, or assignment history is available to determine whether the difference represents workload imbalance.
- **Visual 15 (Listed Nurses by Department):** The departments at the top of the roster chart are tied at four listed nurses. In the full roster, departments have either three or four nurses, so staffing counts are unusually uniform. The chart describes roster distribution only; it does not measure how many patients each nurse served.
- **Visual 16 (Appointment Volume by Facility):** Among the 12 facilities shown, volumes range from 243 at HF017 to 278 at HF013, a spread of 35 appointments. The top facilities are therefore fairly close in absolute appointment counts. The chart omits the other eight facilities and has no capacity denominator, so these counts cannot be read as facility utilization rates.
- **Visual 17 (Bed Roster Status):** All 2,000 listed beds are labeled Available. A single status category contains no variation and should be interpreted as a roster snapshot or source label, not evidence that every bed was unoccupied throughout the year. There is no admission-to-bed link or dated occupancy history.
- **Visual 18 (Equipment Distribution by Department):** The plotted departments each have one equipment record; across the source roster, all 200 departments have exactly one record. The even pattern likely reflects the way this roster was assembled. Equipment counts do not show whether items are functional, available, or used.
- **Visual 19 (Patients per Listed Staff by Department):** The plotted ratios differ across departments because patient counts and roster counts differ, and some departments have small staff denominators. These are crude patients-to-roster comparisons, not assigned patient panels or actual doctor/nurse workload. They should not be used to rank staffing adequacy without staff assignments, shifts, and department capacity data.

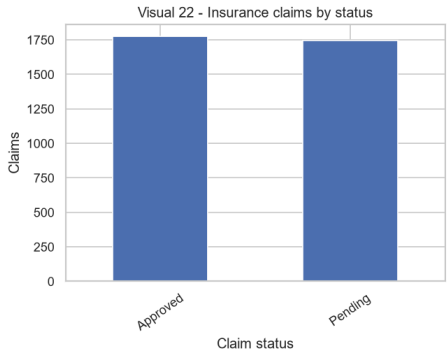
4.4 - Financial Analysis

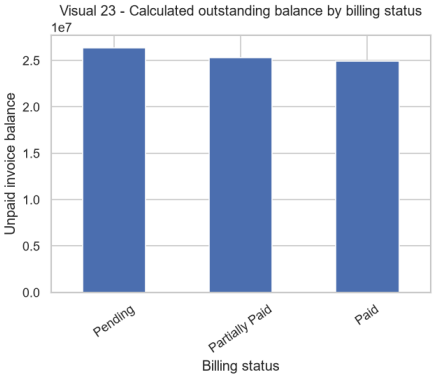
The unlinked monthly `Revenue_Summary` table cannot be joined to appointments, departments, or patients. For linked financial analyses, billing records are connected through `AppointmentID`, payments and insurance claims

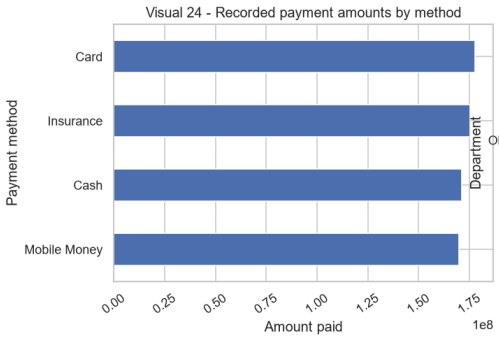
through `BillingID`, and all records are restricted to the master appointment cohort. Payment amounts represent recorded receipts; invoice amounts and outstanding balances are reported separately.

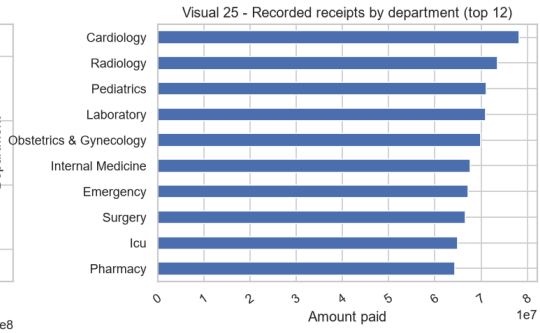













```
Cohort billing records: 5,000; linked claims: 3,521.
Billed amount: 771,133,439.00; recorded payments: 694,551,320.00; calculated outstanding balance: 76,58
2,119.00.
The standalone Revenue_Summary is kept separate for later reconciliation; it is not allocated across de
partments.
```

Interpretation & Financial Insights - Visuals 20 to 25

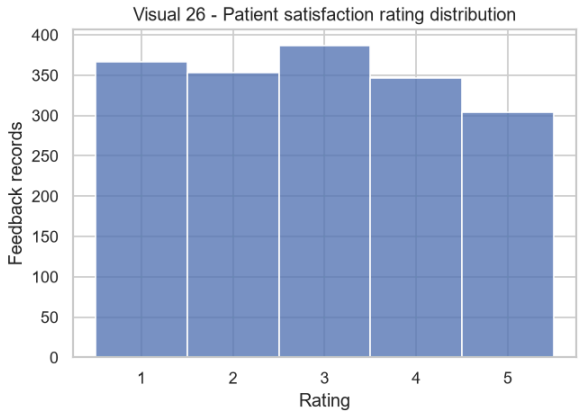
- **Visual 20 (Monthly Billed Amounts and Recorded Receipts):** Billed totals range from 57.2 million in September to 70.2 million in June; linked receipts range from 51.4 million in September to 62.9 million in June. The two lines move in broadly similar directions, and receipts remain below billed amounts in each month. This is a descriptive comparison in source amount units; currency is not specified, and the separate `Revenue_Summary` is not used in this chart.
- **Visual 21 (Billing Distribution by Status):** The 5,000 billing records are split fairly evenly: 1,689 Partially Paid, 1,685 Paid, and 1,626 Pending. The differences are small, so status labels do not reveal a dominant billing state. The next chart comparison is important because the status and payment data are not internally consistent.
- **Visual 22 (Insurance Claims by Status):** The cohort contains 3,521 linked claims: 1,774 Approved and 1,747 Pending. The counts are nearly equal, so approval is not the dominant state in these records. Claim status and claim amount should not be equated with cash received; claims are a separate process linked through BillingID.
- **Visual 23 (Calculated Outstanding Balance by Billing Status):** Calculated balances are similar across the three statuses, from 24.9 million for Paid to 26.4 million for Pending. This is counterintuitive: Paid records still have balances, while Pending records also have large linked payment totals. The plot therefore exposes a status-versus-transaction inconsistency rather than a trustworthy status-based aging profile. Balances use max(Amount - matched AmountPaid, 0) per bill.
- **Visual 24 (Recorded Payment Amounts by Method):** Receipts are spread fairly evenly across Card (177.9 million), Insurance (175.4 million), Cash (171.3 million), and Mobile Money (170.0 million). The 7.9 million difference between the largest and smallest method is modest relative to total receipts; no channel clearly dominates. These totals are transaction amounts, not payment counts or customer preferences.
- **Visual 25 (Recorded Receipts by Department):** Cardiology leads the departments shown at 78.2 million, followed by Radiology at 73.6 million and Pediatrics at 71.1 million; the next departments are close behind. This

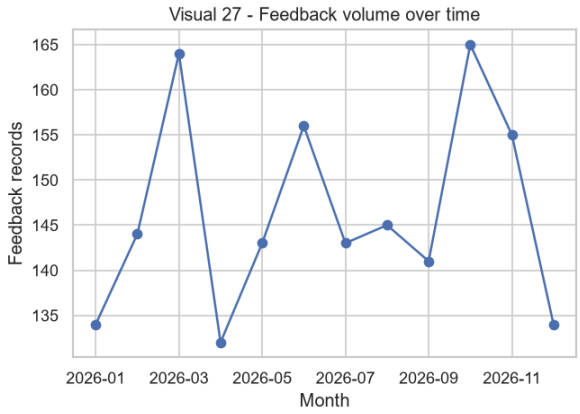
attributes matched payment records to the appointment department. It is not directly comparable with the standalone monthly `Revenue_Summary`, which has no join key and may use a different scope or revenue definition.

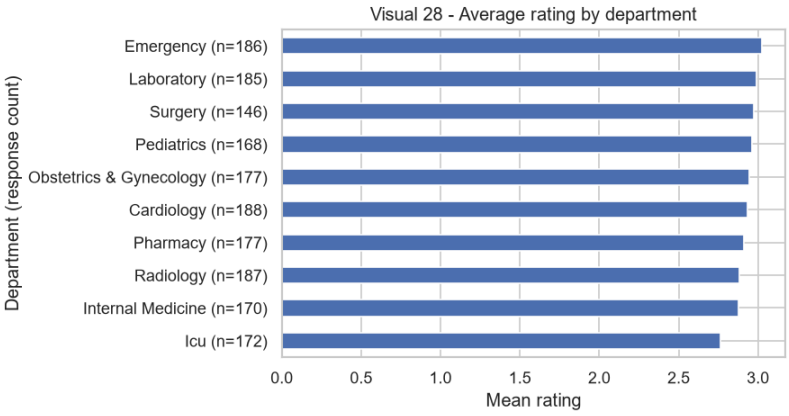
Financial control implication: Do not infer collections or receivables from Billing.Status alone until the source owner reconciles those labels to payment transactions.

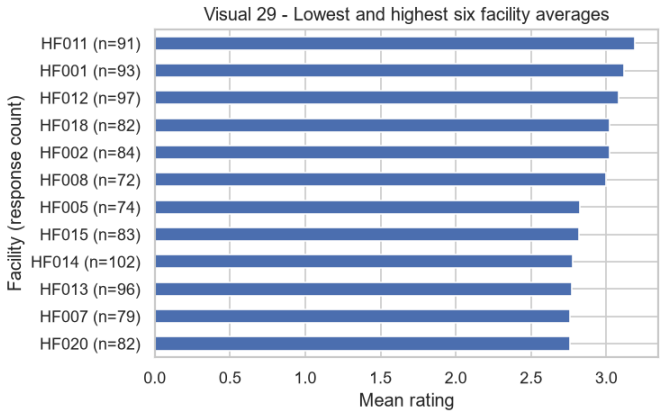
4.5 - Patient Experience

Patient experience is assessed from feedback records linked to the appointment cohort. The analysis reports rating distributions and feedback patterns by month, department, and facility. Feedback coverage is not assumed to represent all patients, and comments are not interpreted as a validated sentiment score.










```
Linked feedback records: 1,756; rated records: 1,756; overall mean rating: 2.93
Monthly feedback counts and mean ratings:
```

month	feedback_count	mean_rating
2026-01	134	3.06
2026-02	144	2.88
2026-03	164	2.90
2026-04	132	2.94
2026-05	143	2.85
2026-06	156	2.99
2026-07	143	2.97
2026-08	145	2.92
2026-09	141	2.82
2026-10	165	3.05
2026-11	155	2.85
2026-12	134	2.90

```
Department rating range: 2.76 to 3.02
Facility range shown in the chart: HF020 (n=82) to HF011 (n=91)
```

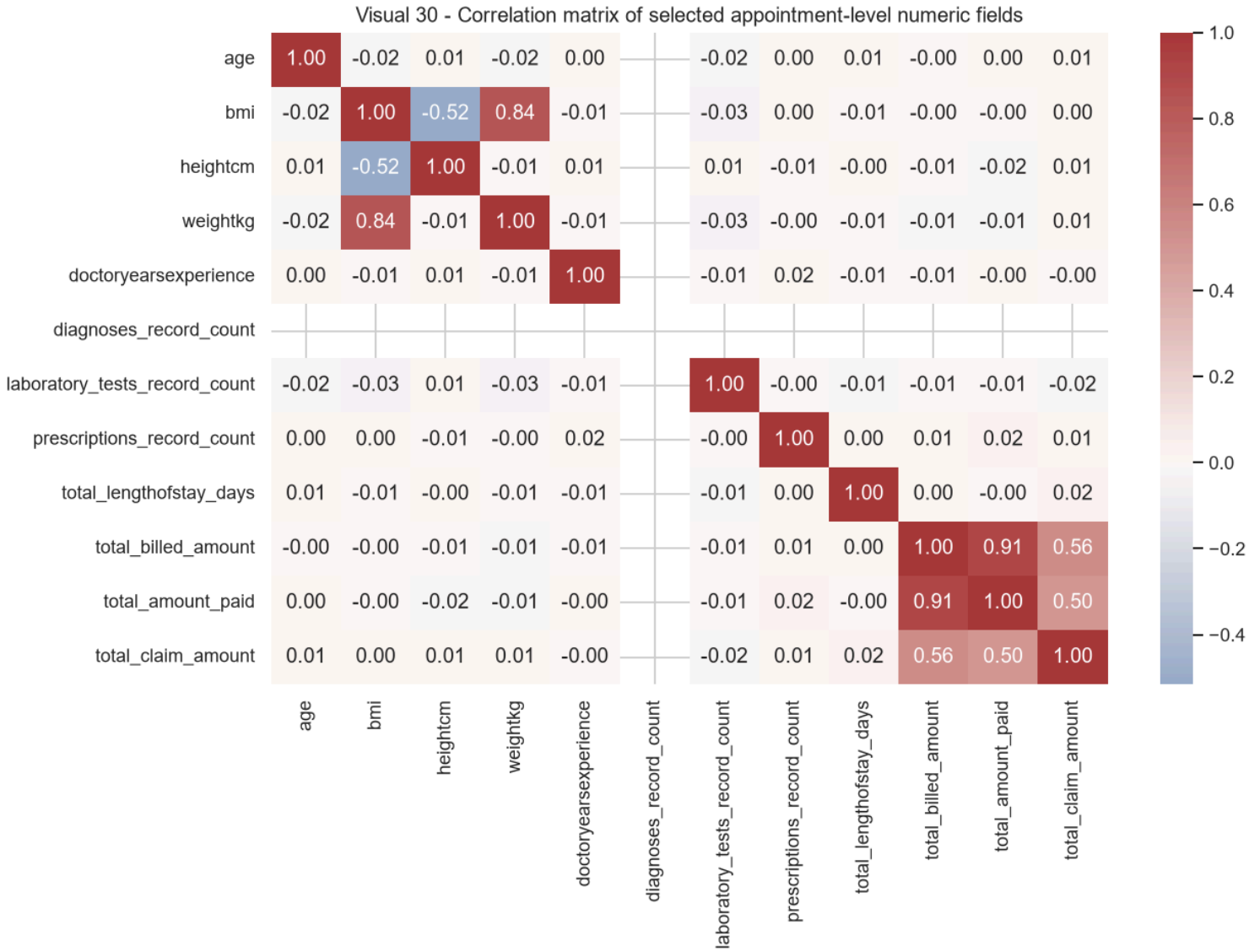
Interpretation & Patient Experience Insights - Visuals 26 to 29

- **Visual 26 (Satisfaction Rating Distribution):** The mode is 3 stars (387 records), followed by 1 star (366), 2 stars (353), 4 stars (346), and 5 stars (304). Ratings are spread across the full scale rather than concentrated at 4-5. In total, 719 responses (40.9%) are ratings 1-2, compared with 650 (37.0%) ratings 4-5; the overall mean is 2.93 and median is 3. This suggests mixed feedback with a slightly larger low-rating share, not uniformly positive experience.
- **Visual 27 (Feedback Volume by Month):** Monthly response counts fluctuate between 132 and 165. October has the most feedback (165) and April the least (132); March is also high at 164. The line shows changing response volume, not a satisfaction trend. Monthly average scores range from 2.82 in September to 3.06 in January and move up and down rather than improving consistently.
- **Visual 28 (Average Rating by Department):** Department averages run from 2.76 in ICU (172 responses) to 3.02 in Emergency (186 responses), a difference of 0.26 on a five-point scale. The departments cluster near three stars; this is a modest separation, not evidence by itself of a major service-quality gap. The response counts are substantial but do not eliminate non-response bias.
- **Visual 29 (Facility Rating Comparison):** The revised chart compares the six lowest and six highest facility averages and labels each with its response count. HF020 is lowest at 2.76 (82 ratings), while HF011 is highest at 3.19 (91 ratings). The visible 0.43-point gap is worth investigating, but facility averages are descriptive and may reflect different patient mix or response patterns; they are not adjusted performance scores.

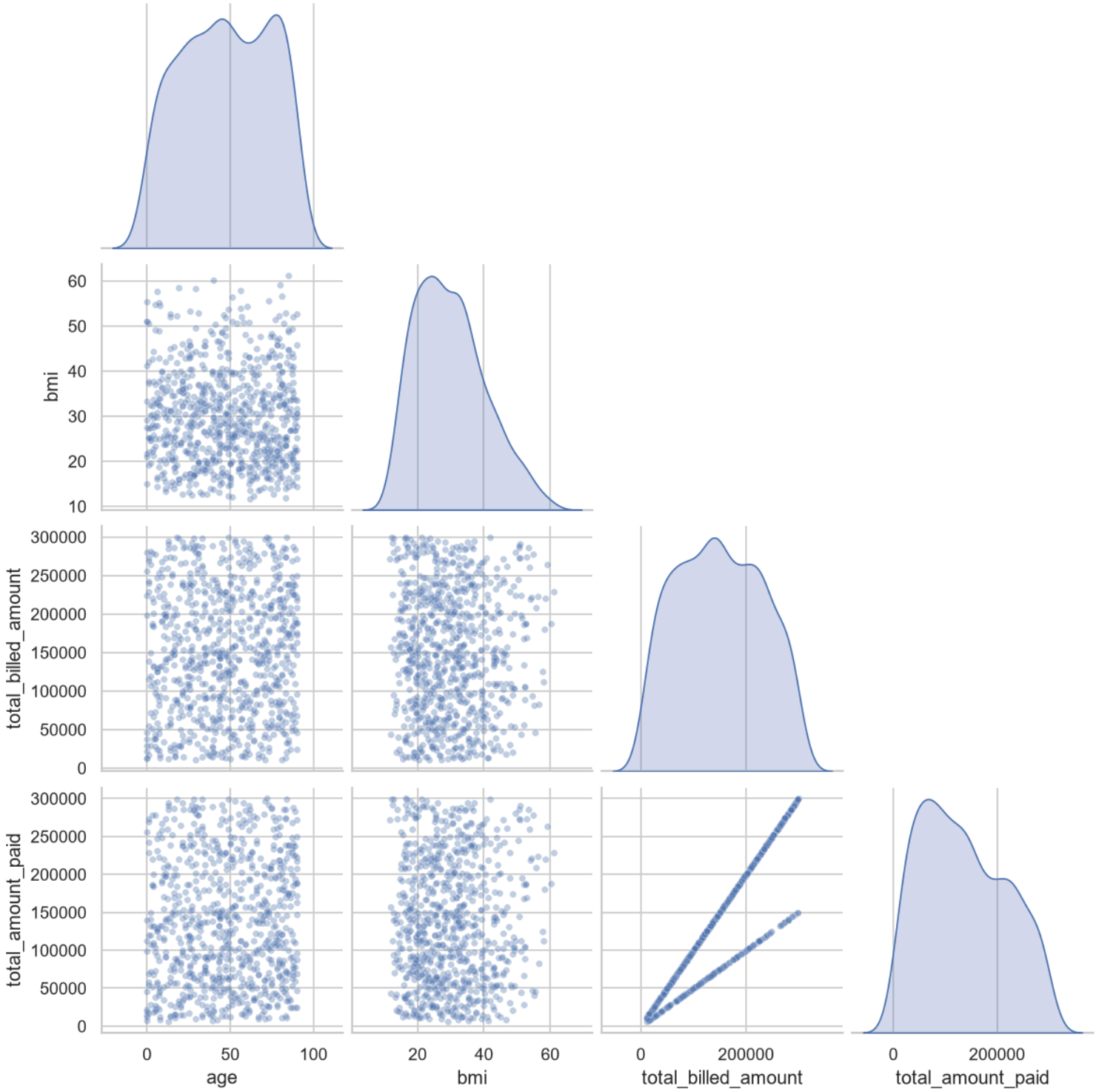
Coverage caveat: Feedback is linked to 1,756 of 5,000 appointments (35.1%). The results summarize submitted feedback and may not represent patients who did not respond.

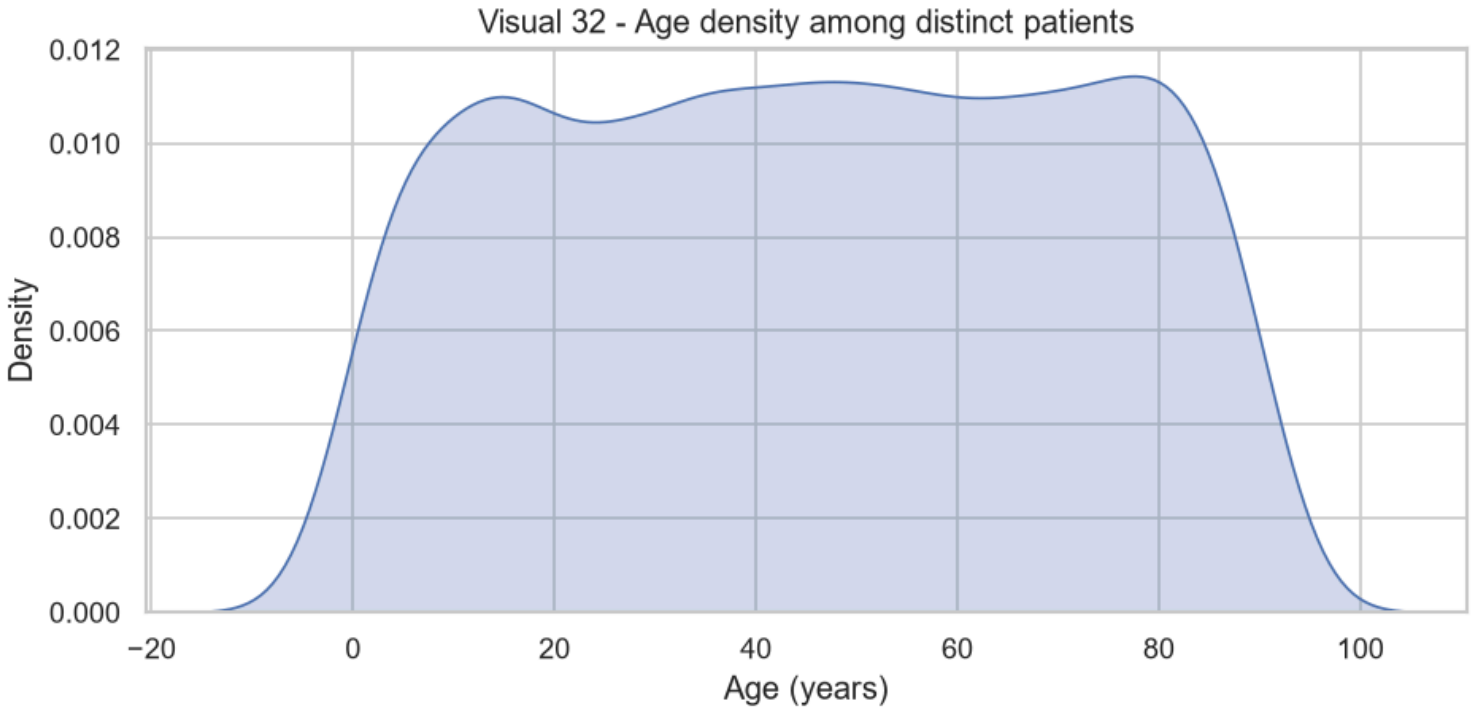
4.6 - Visualization and Multivariate Review

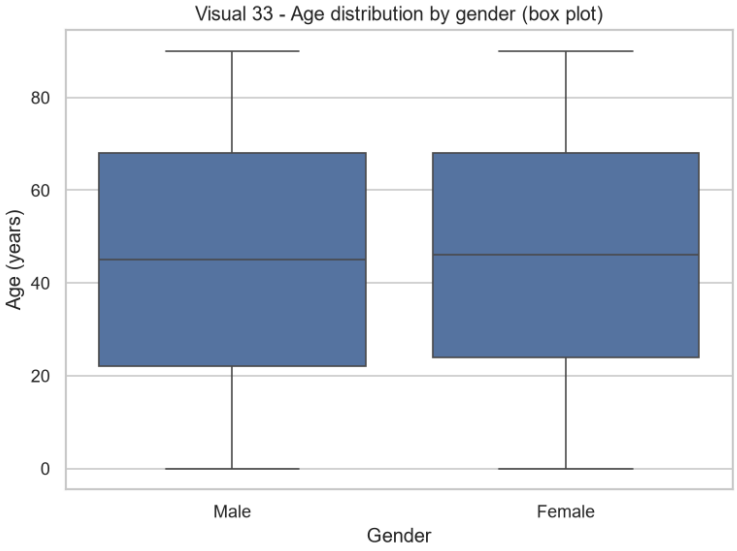
The domain sections above provide histograms, bar charts, and time-series lines. This final review adds a correlation heatmap, a pair plot, a density plot, and a scatter plot for selected numeric fields. Geographic bar charts are used where location labels exist; a true geographic map requires coordinates or boundary geometry that is not part of the provided files.

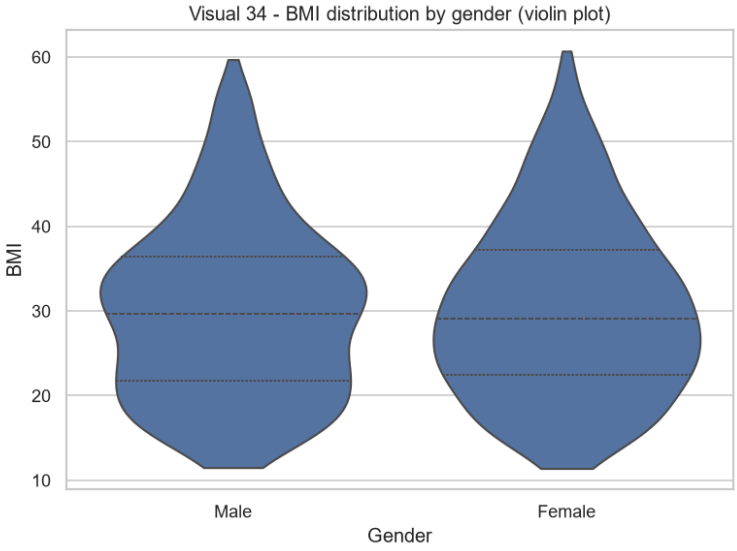


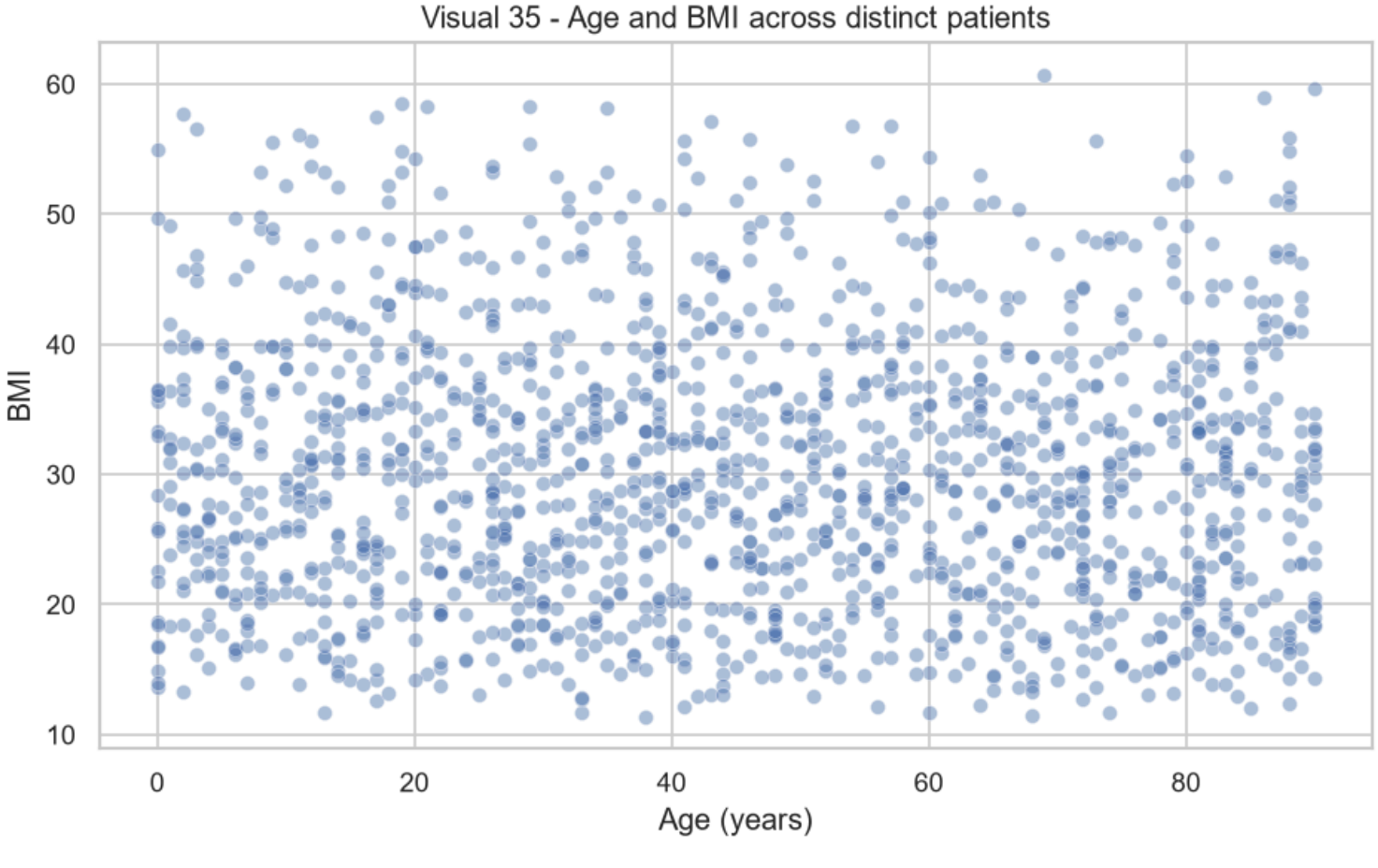
Visual 31 - Pair plot of selected patient and financial variables




Visual 32 - Age density among distinct patients
Density
0.012
0.010
0.008
0.006
0.004
0.002
0.000
−20
0
20
40
60
80
100
Age (years)


Visual 33 - Age distribution by gender (box plot)
Age (years)
80
60
40
20
0
Male
Female
Gender


Visual 34 - BMI distribution by gender (violin plot)
BMI
60
50
40
30
20
10
Male
Female
Gender


Visual 35 - Age and BMI across distinct patients
BMI
60
50
40
30
20
10
0
20
40
60
80
Age (years)

```
Selected numeric variables: ['age', 'bmi', 'heightcm', 'weightkg', 'doctoryearsexperience', 'diagnoses_
record_count', 'laboratory_tests_record_count', 'prescriptions_record_count', 'total_lengthofstay_day
s', 'total_billed_amount', 'total_amount_paid', 'total_claim_amount']
Correlation values are descriptive associations and do not imply causality.
```

Interpretation & Multivariate Insights - Visuals 30 to 35

- **Visual 30 (Correlation Heatmap):** The strongest positive relationships are between BMI and weight (r = 0.84), and between billed amount and paid amount (r = 0.91). BMI and height show a negative association (r = -0.52), which is unexpected for a cohort where BMI is derived from height and weight and deserves a source-formula/data validation check. Billed amount and total claim amount have a moderate positive association (r = 0.56). Most age, clinical-event-count, and length-of-stay correlations with the other fields are close to zero. These are linear associations among appointment-level columns; aggregated financial totals share the same billing/payment process, so their high correlation is not independent evidence of causation.
- **Visual 31 (Pair Plot):** The diagonal panels show the distributions of age, BMI, billed amount, and paid amount; the off-diagonal panels show their pairwise scatter. Age has little visible linear separation from BMI or financial totals. Billed and paid amounts rise together, consistent with the heatmap, but the points do not lie on a single one-to-one line, reflecting partial payments and other transaction differences. The plot uses an 800-row sample for readability and is descriptive, not an inferential test.
- **Visual 32 (Age Density):** The density curve covers a broad age range rather than forming a narrow, bell-shaped peak. There are mild variations in density across age bands, but no single age band accounts for most of the patient cohort. KDE smoothing can make local peaks appear more pronounced; the histogram and summary statistics should be used together.
- **Visual 33 (Age Box Plot by Gender):** Median age is 45 for males and 46 for females, and the interquartile ranges are almost identical (approximately 23-68 and 24-69). The boxes and whiskers overlap heavily, so this cohort shows no material age-profile separation by gender.
- **Visual 34 (BMI Violin Plot by Gender):** BMI distributions overlap closely. Female median BMI is 28.9 (IQR 21.8-36.9) and male median is 29.4 (IQR 21.7-37.3). The similar medians and quartiles suggest little difference in the central BMI distribution between genders; the violin width indicates where observations are more concentrated, not a confidence interval.
- **Visual 35 (Age vs. BMI Scatter Plot):** Points are widely spread vertically across nearly all ages, with no obvious linear slope. Similar BMI values occur at younger and older ages, which agrees with the near-zero age-BMI correlation. This does not rule out nonlinear relationships or clinical effects after adjusting for other factors.

Reading these figures together: The demographic charts count one row per distinct patient, while most heatmap and pair-plot fields are appointment-level aggregates. Repeat appointments and constructed totals can affect apparent associations. Correlation and visual overlap do not establish causality.

Part 5: Feature Engineering

This part creates the ten features requested in the Week 6 brief and adds variables motivated by the EDA. Feature definitions preserve uncertainty in the source. Measures derived during or after a visit must not be used to predict a pre-visit outcome without a historical, leakage-safe reconstruction.

5.1 - Patient, Visit, Clinical, and Financial Features

The following cell derives age groups, a cautious screening marker, visit history, stay bands, insurance-ID availability, revenue quartiles, disease severity, satisfaction, facility feedback scores, and a department workload proxy. Payment indicators are added to investigate the reconciliation issues found in the EDA.

	non-missing rows
appointment_satisfaction_score	1756
high_risk_patient_indicator	4125
previous_appointment_count	5000
patient_age_group	5000
length_of_stay_category	5000
insurance_coverage_category	5000
revenue_category	5000
frequent_visitor_indicator	5000
disease_severity_score	5000
patient_satisfaction_category	5000
facility_performance_score	5000
department_workload_index	5000
payment_collection_ratio	5000
billing_status_payment_conflict	5000

```
Patient Age Group:
```

patient_age_group	rows
65+	1471
0-17	970
18-34	873
35-49	861
50-64	825

```
High Risk Patient Indicator:
```

high_risk_patient_indicator	rows
1	4125
<NA>	875

```
Length Of Stay Category:
```

length_of_stay_category	rows
No admission	4414
Medium stay (4-7 days)	245
Long stay (>7 days)	186
Short stay (1-3 days)	155

```
Insurance Coverage Category:
```

insurance_coverage_category	rows
Insurance ID recorded - coverage unverified	5000

```
Revenue Category:
```

revenue_category	rows
High	1250
Moderate	1250
Very high	1250
Low	1250

```
Disease Severity Score:
```

disease_severity_score	rows
2.0	1696
1.0	1682
3.0	1622

```
Patient Satisfaction Category:
```

patient_satisfaction_category	rows
Not rated	3244
Low (1-2)	719
Moderate (>2-3)	387
High (>3-4)	346
Very high (>4-5)	304

```
Feature table: 5,000 appointments and 4,443 distinct patients.
```

Interpretation & Feature-Engineering Results

- **Age groups:** The 5,000 appointment rows are distributed across all five bands. The largest is age 65+ with 1,471 rows (29.4%); ages 0-17 account for 970 (19.4%). The other groups range from 825 to 873 rows. These are appointment counts, not unique-patient counts; the master contains 4,443 distinct patients.
- **High-risk screening indicator:** It is positive for 4,125 rows and unknown for 875; there are no zero values. In this source, chronic conditions are recorded as named conditions or Not recorded, with no explicit no-condition group. The 82.5% positive share is therefore the result of the indicator definition and available documentation; it must not be read as the prevalence of clinically high-risk patients. Keep the 875 unknowns distinct and consider separating age 65+ from documented chronic history in later models.
- **Admissions and severity:** Of 5,000 appointments, 4,414 (88.3%) have no linked admission; 586 (11.7%) are split into 155 short, 245 medium, and 186 long stays. The severity score covers all rows and is relatively balanced: 1,682 mild (33.6%), 1,696 moderate (33.9%), and 1,622 severe (32.4%). Stay and severity are observed during or after care, so they are not valid pre-visit predictors.
- **Insurance and revenue:** All 5,000 rows have a recorded InsuranceID, but active coverage is unverified, so this feature has no variation for an insured/uninsured comparison. Positive bills form four equal-sized quartiles of 1,250 appointments each; no appointment falls into the No bill group.

- **Satisfaction:** A numeric appointment rating is available for 1,756 rows (35.1%); 3,244 (64.9%) remain Not rated. Among rated rows, 719 are Low (41.0%), 387 Moderate (22.0%), 346 High (19.7%), and 304 Very high (17.3%). The limited response coverage means the rating profile may not represent all patients.
- **Facility and department measures:** The facility score and workload index are mapped to all 5,000 appointment rows. The former is a facility-level average based on feedback, while the latter is a department percentile based on listed staff, not actual shift workload. Part 3's facility mapping mismatch limits facility attribution.
- **Payment features:** The collection ratio and status-conflict flag make payment reconciliation visible for subsequent analysis. Because a transaction can exceed its bill and source status labels conflict with payments, these values describe data records and should be reviewed before finance uses them as performance indicators.

5.2 - Feature Dictionary and Modeling Readiness

This dictionary documents feature definition, grain, and timing so future models can avoid target leakage and distinguish measured values from proxies.

	Feature	Definition	Grain	Modeling timing and caution
0	patient_age_group	Five age bands from 0-17 through 65+	Appointment	Available at encounter
1	high_risk_patient_indicator	Age 65+ or documented chronic condition	Appointment	Screening proxy; unknown history stays missing
2	previous_appointment_count	Earlier dated appointments for the same patient	Appointment	Chronology-safe candidate
3	frequent_visitor_indicator	1 from the third observed appointment onward	Appointment	Chronology-safe; analyst-defined threshold
4	length_of_stay_category	No admission, short, medium, long, or not assessable	Appointment	Post-admission outcome; leakage risk
5	insurance_coverage_category	Insurance ID recorded or absent	Appointment	Active coverage is unverified
6	revenue_category	Quartile of positive appointment bills or no bill	Appointment	Post-encounter outcome; leakage risk
7	disease_severity_score	Mild=1, Moderate=2, Severe=3	Appointment	Clinical outcome; leakage risk
8	facility_performance_score	Mean linked rating with at least 30 responses	Facility	Use historical/out-of-fold values
9	patient_satisfaction_category	Rating band or not rated	Appointment	Outcome; not a same-visit predictor
10	department_workload_index	Percentile rank of appointments per listed staff	Department	Roster proxy; historical values needed
11	payment_collection_ratio	Recorded payments divided by billed amount	Appointment	Post-encounter financial measure
12	billing_status_payment_conflict	Flag for status/payment disagreement	Appointment	Quality indicator

Interpretation & Feature Dictionary

The dictionary contains the ten requested engineered variables plus prior-visit history and two financial quality indicators. The coverage table shows that most derived fields are populated for all 5,000 appointment rows. The meaningful exceptions are the high-risk marker, which has 4,125 positive values and 875 unknowns with no confirmed low-risk cases, and the numeric satisfaction score, available for only 1,756 appointments. The satisfaction category retains all 3,244 missing-feedback appointments as Not rated rather than imputing a score.

The table also clarifies timing: prior appointment count is the clearest chronology-safe feature. Stay duration, disease severity, billed revenue, satisfaction, payment ratio, and same-period facility/workload aggregates are measured during or after the encounter. They describe the cohort but would leak information into a pre-visit model unless rebuilt from earlier periods. Insurance-ID presence is constant in this extract and is not a measure of active coverage.

Part 6: Statistical Analysis & Hypothesis Testing

This part reports descriptive statistics, the requested relationships, a correlation heatmap, and formal tests. Each result states its observation grain so repeated visits, department aggregates, and monthly observations are interpreted appropriately.

6.1 - Descriptive Statistics

The summary includes N, mean, median, mode, standard deviation, variance, quartiles, and percentiles. Age is summarized once per patient, financial and encounter measures at appointment grain, and satisfaction from rated records only.

Measure	N	Mean	Median	Mode	Std. deviation	Variance	Q1 / P25	Q3 / P75	P10	
Patient age (one row per patient)	4443	45.518	46.00	36.0	26.149	6.837950e+02	23.000	68.00	9.0	
Length of stay (appointments; zero means no admission)	5000	0.664	0.00	0.0	2.067	4.272000e+00	0.000	0.00	0.0	
Billed amount (appointment rows)	5000	154226.688	153724.00	13769.0	83630.750	6.994102e+09	81148.750	226674.75	39020.9	270
Recorded payments (appointment rows)	5000	138910.264	128223.50	13769.0	83367.278	6.950103e+09	66926.500	209092.50	31461.3	263
Disease severity score (ordinal 1-3)	5000	1.988	2.00	2.0	0.813	6.610000e-01	1.000	3.00	1.0	
Appointment satisfaction rating (feedback only)	1756	2.925	3.00	3.0	1.386	1.920000e+00	2.000	4.00	1.0	
Department workload index (one row per department)	200	50.250	50.25	14.5	28.924	8.365810e+02	26.375	75.25	10.5	

Interpretation of descriptive statistics. The age summary uses 4,443 distinct patients: the mean is 45.5 years, the median 46, and the middle half spans 23 to 68 years. The first and third quartiles are far apart, showing a broad age profile rather than a cohort concentrated around one age.

Length of stay is summarized across all 5,000 appointments, including 4,414 with no admission. Its median and third quartile are both zero, and the 90th percentile is two days; these values are dominated by non-admissions and do not describe duration among hospitalized patients. The 586 admission-linked encounters should be analyzed separately for typical stay length.

Billed amounts average 154,226.69 source units (median 153,724), while recorded payments average 138,910.26 (median 128,223.50). Their standard deviations are both about 83,000, and their 10th-to-90th percentile ranges are wide. This spread supports using medians and quantiles alongside means; the units are not identified as a currency. Disease severity is nearly balanced across its three levels, so its mean of 1.99 is less informative than the category counts.

Among 1,756 rated appointments, satisfaction averages 2.925/5, with median and mode 3 and an interquartile range of 2 to 4. The standard deviation of 1.386 indicates ratings vary substantially around the middle score. The department workload index is centered at 50.25 across 200 departments; its 10th and 90th percentiles are 10.5 and 89.175, respectively, showing substantial variation in the roster-based proxy.

6.2 - Correlation Analysis and Heatmap

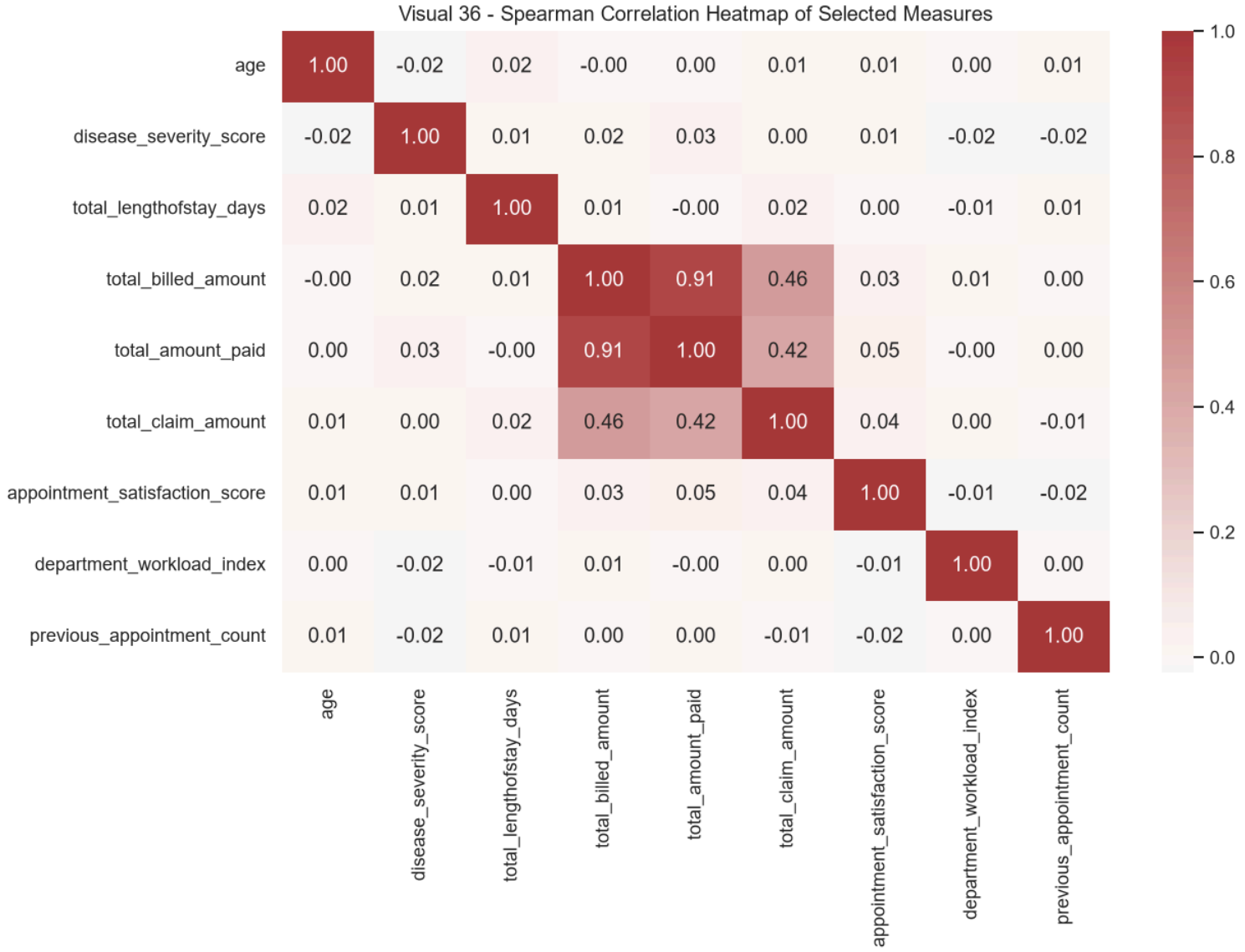
The analysis addresses all requested relationships using patient, admitted-patient, department, or month observations. Insurance coverage versus utilization is reported as not estimable when the extract has no verified coverage variation.

	Relationship	Method	N	Coefficient	P-value	Note
0	Age vs admission at first observed appointment	Point-biserial	4443	0.0255	0.0893	One first appointment per patient; admission is binary.
1	Age vs billing at first observed appointment	Spearman	4443	-0.0067	0.6568	One first appointment per patient; billing may be skewed.
2	Diagnosis severity vs length of stay among admitted patients	Spearman	578	-0.0169	0.6843	First admitted encounter per patient; severity is ordinal.
3	Department workload index vs mean satisfaction	Spearman	200	-0.0245	0.7310	Department grain; roster-based workload proxy.
4	Monthly linked receipts vs distinct patients	Spearman	12	0.9161	0.0000	Monthly observations; not a forecasting test.

insurance_coverage_category	distinct_patients	appointments	mean_prior_visits
Insurance ID recorded - coverage unverified	4443	5000	0.1214

```
Coverage-utilization comparison is not estimable: active coverage is unverified and no second Insurance
ID group is present.
Spearman correlation matrix used in the heatmap:
```

	age	disease_severity_score	total_lengthofstay_days	total_billed_amount	total_am
age	1.00	-0.02	0.02	-0.00	
disease_severity_score	-0.02	1.00	0.01	0.02	
total_lengthofstay_days	0.02	0.01	1.00	0.01	
total_billed_amount	-0.00	0.02	0.01	1.00	
total_amount_paid	0.00	0.03	-0.00	0.91	
total_claim_amount	0.01	0.00	0.02	0.46	
appointment_satisfaction_score	0.01	0.01	0.00	0.03	
department_workload_index	0.00	-0.02	-0.01	0.01	
previous_appointment_count	0.01	-0.02	0.01	0.00	



Interpretation of correlations and heatmap. Four of the five requested pairwise relationships are close to zero in this extract. Age and admission at the first observed encounter have a point-biserial coefficient of 0.0255 (p = 0.0893; n = 4,443). Age and first-visit billed amount have Spearman ρ = -0.0067 (p = 0.6568). Among 578 admitted patients, diagnosis severity and stay duration have ρ = -0.0169 (p = 0.6843). Across 200 departments, workload index and mean satisfaction have ρ = -0.0245 (p = 0.7310). These results do not show meaningful monotonic or linear associations in the selected samples.

The exception is monthly linked receipts versus distinct patient volume: Spearman ρ = 0.9161 (p < 0.001; n = 12 months). Higher-volume months tended to have higher recorded receipts. This is a strong association over only one

year of observations, not evidence of causation or a validated forecasting relationship.

The heatmap confirms the strongest pair is billed amount versus recorded payments (ρ = 0.91). Billed amount is also moderately associated with claim amount (ρ = 0.46), and payments with claim amount (ρ = 0.42). All other displayed pairs are close to zero (absolute ρ no greater than 0.05 at the shown precision). The financial fields are linked parts of the same billing process, so their associations are expected and do not imply that claims cause payment. Insurance utilization cannot be compared across coverage groups: all 4,443 patients have an InsuranceID recorded, covering 5,000 appointments, and active coverage status is unavailable. The heatmap is descriptive and combines variables at different analytical grains.

6.3 - Hypothesis Testing

Five feasible tests are conducted. Each result states H0, H1, test, significance level, p-value, sample size, effect size where applicable, and conclusion. Holm adjustment controls family-wise error at α = 0.05. Comparisons requiring a verified uninsured group are marked not estimable.

	Hypothesis	H0	H1	Test	Statistic	P-value	N	Effect size	Note	Holm adjusted p	D al
0	H1: Satisfaction differs across facilities	Rating distributions are the same across facilities.	At least one facility rating distribution differs.	Kruskal-Wallis	14.93140	0.72695	1675	0.00000	20 facilities; epsilon-squared effect size.	1.00000	
1	H2: Admission rate differs by age group	Admission and age group are independent.	Admission rate differs across age groups.	Chi-square independence (df=4)	9.75209	0.04482	4443	0.04685	Cramer's V; minimum expected cell=85.1.	0.17927	
2	H3: Severity is associated with stay duration	Severity and stay duration have no monotonic association.	Severity and stay duration have a monotonic association.	Spearman rank correlation	-0.01695	0.68432	578	0.01695	Admitted cohort; absolute rho is effect magnitude.	1.00000	
3	H4: Workload is associated with satisfaction	Workload and mean satisfaction have no monotonic association.	Workload and mean satisfaction have a monotonic association.	Spearman rank correlation	-0.02446	0.73102	200	0.02446	Department grain; workload is a roster proxy.	1.00000	
4	H5: Receipts are associated with patient volume	Monthly receipts and patient volume have no monotonic association.	Monthly receipts and patient volume have a monotonic association.	Spearman rank correlation	0.91608	0.00003	12	0.91608	Monthly points; not a forecasting or causal test.	0.00014	

patient_age_group	Admission rate (%)
0-17	10.0
18-34	11.6
35-49	9.6
50-64	13.8
65+	12.4

```
No pairwise facility follow-up: the global test does not survive Holm adjustment.
Insured-versus-uninsured billing test is not estimable: active status is unverified and no uninsured gr
oup is present.
```

Interpretation & Statistical Conclusions. The five primary tests use α = 0.05 and Holm-adjusted p-values. The final decision is based on the adjusted value, while the raw p-value and effect size help show the strength and practical scale of each result.

- **Satisfaction across facilities:** Kruskal-Wallis gives H = 14.931, raw p = 0.72695, and adjusted p = 1.000 across 1,675 first-rated patient encounters and 20 facilities. Epsilon-squared is approximately 0. The test does not detect different rating distributions. Facility-level interpretation is additionally limited by the 94.9% DepartmentID/FacilityID mismatch found in Part 3
- **Admission by age group:** The chi-square statistic is 9.752 with 4 degrees of freedom (p = 0.04482; adjusted p = 0.17927; n = 4,443). Rates range from 9.6% in ages 35-49 to 13.8% in ages 50-64; the other groups are 10.0%, 11.6%, and 12.4%. Cramer's V = 0.0469 indicates a very small association. Although the unadjusted p-value is below 0.05, the result does not survive Holm correction, so the adjusted decision is to fail to reject independence
- **Severity and stay duration:** Among 578 admitted patients, Spearman ρ = -0.01695 (p = 0.68432; adjusted p = 1.000). The estimated relationship is negligible and not statistically significant
- **Department workload and satisfaction:** Across 200 departments, ρ = -0.02446 (p = 0.73102; adjusted p = 1.000). This provides no evidence of a monotonic association in the department-level data; workload is only a roster proxy
- **Monthly receipts and patient volume:** Across 12 monthly observations, ρ = 0.91608 (p = 0.00003; adjusted p = 0.00014). This is the only test that rejects H0 after correction. Months with more distinct patients tended to have higher receipts, but the short series and shared billing process prevent causal or long-term claims

The insured-versus-uninsured billing test remains unestimable because active coverage is unknown and no uninsured comparison group is present. A failure to reject H0 is not proof that groups are identical.

6.4 - Statistical Analysis Summary

The descriptive table shows a broad patient-age distribution, highly variable financial amounts, and satisfaction centered at 3 among the 1,756 rated appointments. The correlation analyses find near-zero relationships for age, severity/stay, and department workload/satisfaction. The one clear association is monthly receipts with patient volume (ρ = 0.916 over 12 months). Among the five hypothesis tests, this monthly relationship alone remains significant after Holm correction; age-group admission has a raw p below 0.05 but an adjusted p of 0.179 and a very small effect (Cramer's V = 0.047).

These results are descriptive of the available extract. They are constrained by the 35.1% feedback coverage, unverified insurance status, billing/payment conflicts, the 94.9% department/facility mapping mismatch, and only 12 monthly observations. Resolve source issues and collect additional periods before treating facility comparisons, financial trends, or monthly associations as operational forecasts.

Part 7: Predictive Analytics, Forecasting & Clustering

This part develops baseline forecasts for patient demand and linked financial activity, classification models for admission and patient satisfaction, and clustering analyses for patients, facilities, and departments. The available extract contains one year of appointments, so forecasts are deliberately treated as short-data baselines rather than reliable long-term projections. Predictive features are selected to avoid using direct outcome fields as inputs.

7.1 - Patient Volume, Revenue, and Department Workload Forecasts

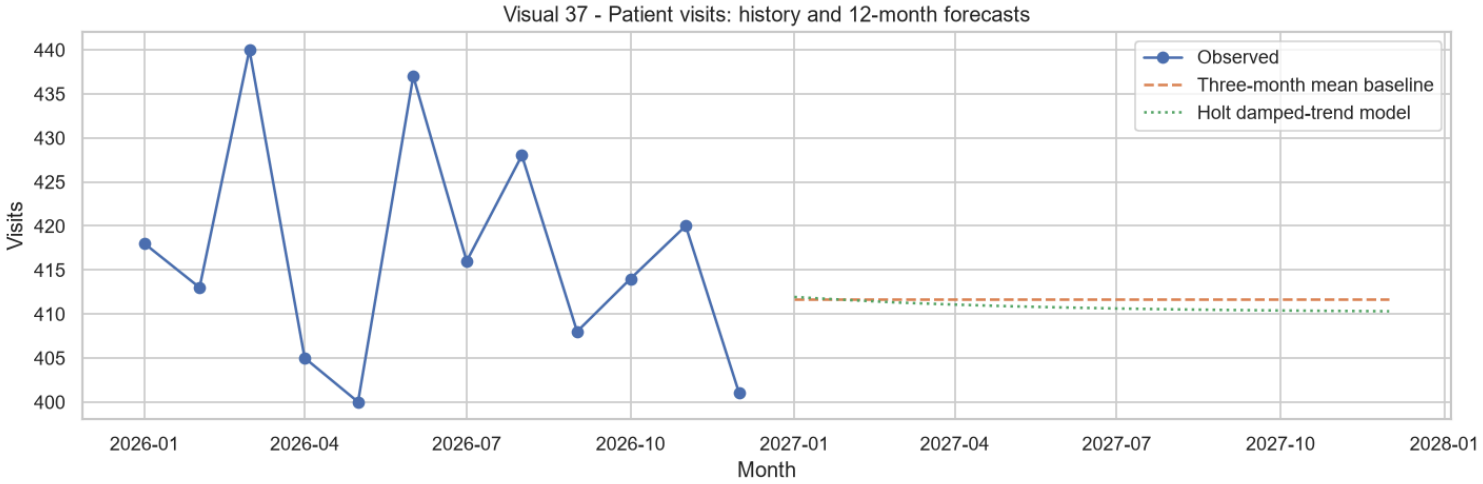
The assignment requests 12-month forecasts for patient visits, hospital revenue, and department demand. Linked appointment bills and payments are used instead of the unlinked Revenue_Summary table. The code compares a three-month moving-average baseline with Holt's damped-trend exponential-smoothing model, then uses a rolling one-step backtest on the final three observed months. Annual seasonality is not fitted because only 12 months are available. Department forecasts remain a simple recent-average baseline because each department series is sparse.

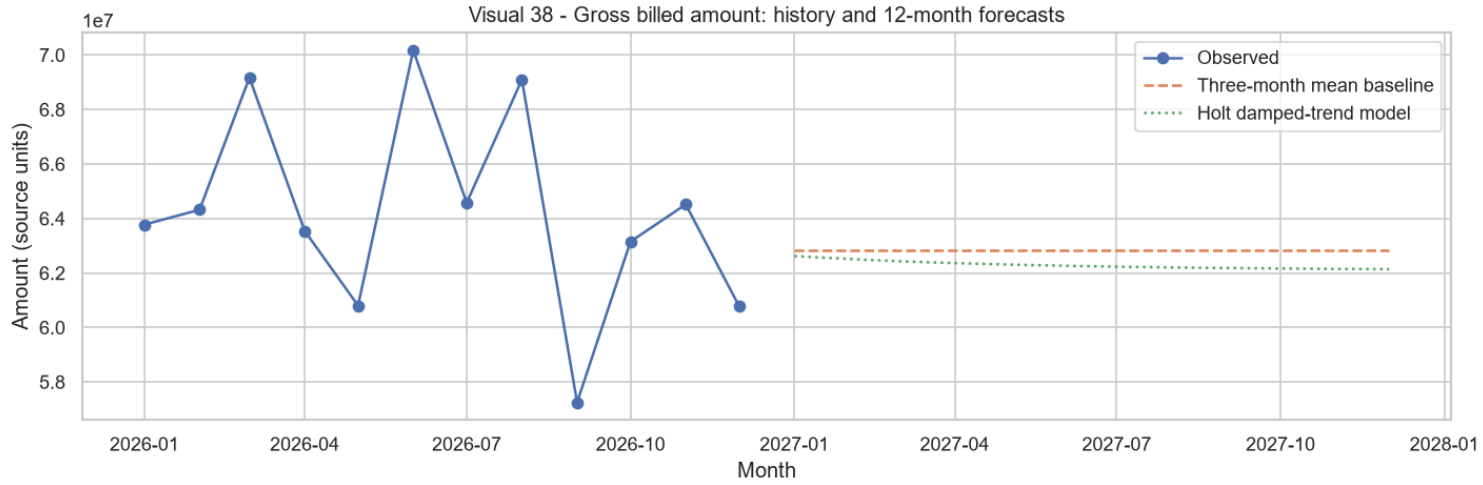
Measure	Model	Backtest MAE (3 months)	Backtest RMSE (3 months)
Patient visits	**Three-month mean baseline**	6.56	7.98
	Holt damped-trend model	7.29	9.69
Gross billed amount	**Three-month mean baseline**	900199.00	967053.73
	Holt damped-trend model	1600514.24	1824504.70
Recorded receipts	**Three-month mean baseline**	1542975.22	1830604.39
	Holt damped-trend model	1819091.83	2223476.77

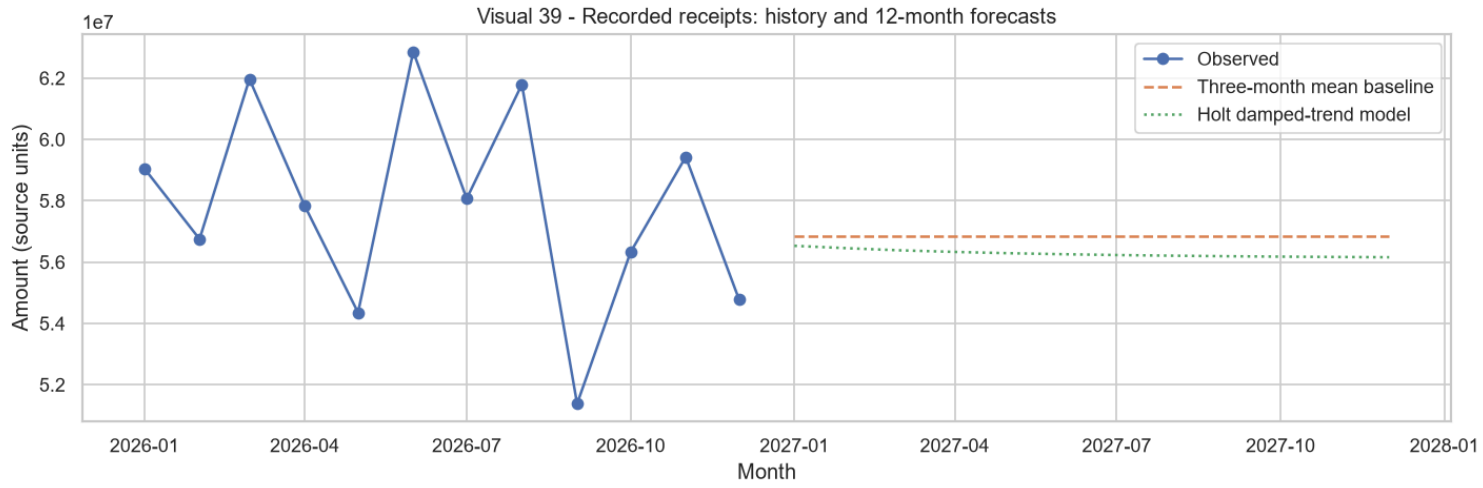
	Patient visits		Gross billed amount		Recorded receipts	
	Three-month mean baseline	Holt damped-trend model	Three-month mean baseline	Holt damped-trend model	Three-month mean baseline	Holt damped-trend model
Forecast month						
2027-01-01	411.67	411.92	62813995.0	62617212.47	56837469.0	56526936.11
2027-02-01	411.67	411.57	62813995.0	62511856.09	56837469.0	56445458.35
2027-03-01	411.67	411.29	62813995.0	62427570.99	56837469.0	56380276.13
2027-04-01	411.67	411.06	62813995.0	62360142.91	56837469.0	56328130.36
2027-05-01	411.67	410.88	62813995.0	62306200.45	56837469.0	56286413.75
2027-06-01	411.67	410.73	62813995.0	62263046.48	56837469.0	56253040.45
2027-07-01	411.67	410.61	62813995.0	62228523.30	56837469.0	56226341.82
2027-08-01	411.67	410.52	62813995.0	62200904.76	56837469.0	56204982.91
2027-09-01	411.67	410.45	62813995.0	62178809.93	56837469.0	56187895.79
2027-10-01	411.67	410.39	62813995.0	62161134.06	56837469.0	56174226.09
2027-11-01	411.67	410.34	62813995.0	62146993.37	56837469.0	56163290.33
2027-12-01	411.67	410.30	62813995.0	62135680.81	56837469.0	56154541.72

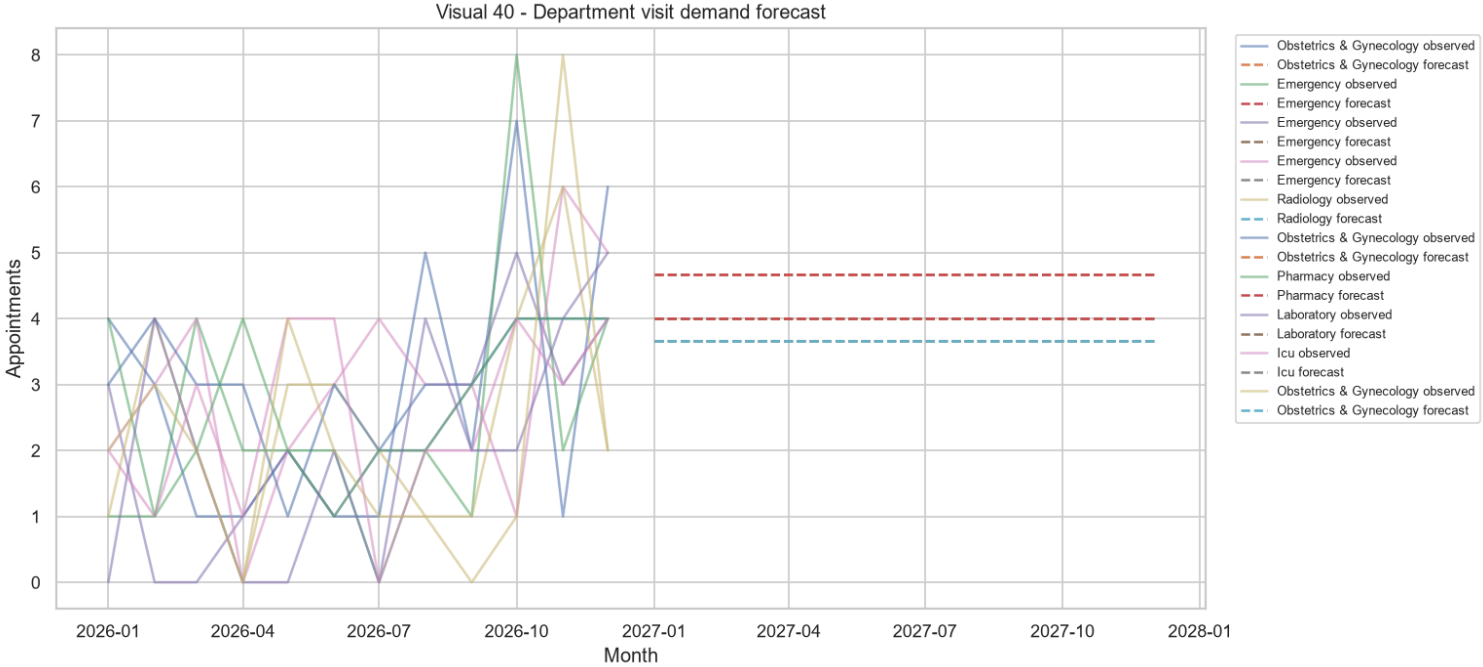
```
Departments with the highest baseline demand over the next 12 months:
```

	departmentid	departmentname	forecast_visits
114	DEP115	Obstetrics & Gynecology	56.0
100	DEP101	Emergency	56.0
80	DEP081	Emergency	48.0
180	DEP181	Emergency	48.0
155	DEP156	Radiology	48.0
94	DEP095	Obstetrics & Gynecology	48.0
97	DEP098	Pharmacy	48.0
26	DEP027	Laboratory	44.0
88	DEP089	Icu	44.0
54	DEP055	Obstetrics & Gynecology	44.0


Visual 37 - Patient visits: history and 12-month forecasts
Observed
Three-month mean baseline
Holt damped-trend model
Visits
Month


Visual 38 - Gross billed amount: history and 12-month forecasts
1e7
Observed
Three-month mean baseline
Holt damped-trend model
Amount (source units)
Month


Visual 39 - Recorded receipts: history and 12-month forecasts
1e7
Observed
Three-month mean baseline
Holt damped-trend model
Amount (source units)
Month


Visual 40 - Department visit demand forecast
Appointments
Month
Obstetrics & Gynecology observed
Obstetrics & Gynecology forecast
Emergency observed
Emergency forecast
Radiology observed
Radiology forecast
Pharmacy observed
Pharmacy forecast
Laboratory observed
Laboratory forecast
Icu observed
Icu forecast

Visuals 37 to 40. The chart panels compare the observed series with both 12-month forecasting approaches.

Forecasting theory and results. The three-month moving average is a statistical baseline: for every future horizon h it uses the same estimate,

$$\hat{y}_{t+h} = (y_t + y_{t-1} + y_{t-2})/3.$$

It is simple to interpret, but assumes neither trend nor seasonality. For the 200 department series, monthly records are sparse, so the same recent-average method is used as a planning baseline.

Holt's damped-trend exponential smoothing is the statistical forecasting model used for the overall monthly series. It updates an estimated level L and trend T from each new observation y. The level gives weight α to the newest value, while the trend uses β to update its slope; φ dampens that slope in future months:

$$L_t = \alpha y_t + (1-\alpha)(L_{t-1} + \varphi T_{t-1}), \quad T_t = \beta(L_t - L_{t-1}) + (1-\beta)\varphi T_{t-1}.$$

The forecast is the current level plus a progressively damped trend. The code uses fixed parameters α = 0.3, β = 0.1, and φ = 0.8; these are transparent modelling choices, not values selected by a large training history. No annual seasonal component is fitted: one year of monthly values is not enough to estimate a repeating yearly pattern reliably.

The backtest makes rolling one-month-ahead predictions for the final three observed months, using only earlier months at each step. MAE is the mean absolute forecast error; RMSE gives more weight to large errors. Lower values are preferable, but three test months are far too few to establish reliable model superiority, and this backtest does not validate a 12-month horizon.

On this short backtest, the three-month mean has lower errors for all three measures. For patient visits, MAE/RMSE are 6.56/7.98 versus 7.29/9.69 for Holt. For gross billed amount, they are 900,199/967,054 versus 1,600,514/1,824,505 source units. For recorded receipts, they are 1,542,975/1,830,604 versus 1,819,092/2,223,477. The evidence therefore favors the simpler baseline for this limited comparison, not as a proven long-term winner.

The moving-average forecast is 411.67 visits per month, 62.81 million in gross bills, and 56.84 million in recorded receipts. Holt starts at about 411.92 visits, 62.62 million billed, and 56.53 million in receipts in January 2027, then declines gently through the year as its damped trend approaches a stable level. The department baseline ranks DEP115 (Obstetrics & Gynecology) and DEP101 (Emergency) highest at 56 visits each over the next 12 months. These values are scenarios in source units, not commitments. Billed totals are not verified accounting revenue, the currency is unspecified, and billing-status/payment conflicts remain unresolved.

7.2 - Hospital Admission Prediction

This model estimates whether an encounter is linked to an admission. It keeps the first observed encounter per patient and uses a stratified 75/25 train/test split. Logistic regression models the log-odds as a weighted combination of input variables,

$$\log(p/(1-p)) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k,$$

then converts the result to an admission probability between 0 and 1. Numeric features are median-imputed and standardized, while categorical features are mode-imputed and one-hot encoded. Class weighting gives the minority admission class more influence during fitting.

Predictors include demographics, visit type, department, chronic-condition history, prior visits, diagnoses, and severity. Diagnosis and severity are assumed available when supporting an admission decision. Billing, length of stay, feedback, and the admission target are excluded to avoid direct target leakage.

	Test rows	Admission prevalence	Accuracy	Balanced accuracy	ROC AUC
0	1111	0.115	0.515	0.468	0.466

```
Classification report:
```

	precision	recall	f1-score	support
0	0.872	0.529	0.659	983.000
1	0.101	0.406	0.162	128.000
accuracy	0.515	0.515	0.515	0.515
macro avg	0.487	0.468	0.410	1111.000
weighted avg	0.784	0.515	0.601	1111.000

```
Confusion matrix (rows = actual, columns = predicted; class order 0, 1):
```

	Predicted no admission	Predicted admission
Actual no admission	520	463
Actual admission	76	52

Interpretation of the admission model. The test set contains 1,111 unique patients, of whom 128 (11.5%) had an admission on their first observed encounter. The logistic model reaches 51.5% accuracy, 46.8% balanced accuracy, and ROC AUC 0.466. The admission recall is 40.6%, but precision is only 10.1%: of 515 encounters predicted as admissions, just 52 were actual admissions, while 463 were false positives. It also misses 76 of the 128 actual admissions.

These results do not support using this model for admission decisions. The AUC is below 0.5 and balanced accuracy is below chance-level 0.5; the class-weighted model produces many false alerts without reliably separating admissions. The first-encounter holdout avoids putting the same patient in both sets, but it is a random split rather than a future-time evaluation. More informative clinical and encounter data, a temporal validation set, calibration, and clinical review would be needed before reconsideration. The model is an educational baseline, not a clinical decision tool.

7.3 - Patient Satisfaction Prediction

This model predicts one of the five recorded rating classes (1-5) from demographic, clinical, and appointment characteristics. It uses the first rated encounter per patient and a stratified 75/25 train/test split. A multiclass logistic regression estimates a score for each rating class and converts the scores to class probabilities with the softmax function; the most probable class is selected. Numeric inputs are imputed and standardized, while categorical inputs are imputed and one-hot encoded.

The target rating and post-visit financial outcomes are excluded from predictors. This analysis applies only to patients who submitted feedback; the EDA found linked ratings for 1,756 of 5,000 appointments (35.1%).

	Feedback records	Test rows	Accuracy	Balanced accuracy
0	1675	419	0.203	0.201

```
Rating-level classification report:
```

	precision	recall	f1-score	support
1	0.217	0.149	0.177	87.000
2	0.256	0.262	0.259	84.000
3	0.214	0.261	0.235	92.000
4	0.192	0.181	0.186	83.000
5	0.133	0.151	0.141	73.000
accuracy	0.203	0.203	0.203	0.203
macro avg	0.202	0.201	0.200	419.000
weighted avg	0.205	0.203	0.202	419.000

Interpretation of the satisfaction model. After retaining one first-rated encounter per patient, the data contain 1,675 patients; 419 are in the test set. Accuracy is 20.3% and balanced accuracy is 20.1%, almost the 20% accuracy expected from chance-level guessing across five equally considered classes. Class-level F1 scores range from 0.141 to 0.259, with no rating category predicted reliably.

The available demographic, clinical, and appointment fields do not provide useful out-of-sample rating prediction in this baseline. Feedback is also selective: 35.1% of appointment rows have a linked rating, so respondents may differ from non-respondents. The model should not be used to rank departments, facilities, or staff. Improving response coverage and investigating low ratings directly are more actionable than acting on these predictions.

7.4 - Patient, Facility, and Department Clustering

K-means assigns each observation to the nearest centroid and minimizes the total squared distance from observations to their assigned centroids. Since distance depends on measurement scale, the variables are standardized first. The patient analysis compares 2-5 cluster solutions using silhouette score; larger values indicate better separation in this feature space. Facility and department analyses use three clusters for a compact operational profile, so those group counts are descriptive choices rather than an optimized result.

Patient inputs summarize age, chronic-history documentation, appointment frequency, mean severity, and satisfaction; admission count is reported after clustering for profile comparison, not used to form the patient groups. Missing numeric values are median-imputed for the clustering step. Facility and department inputs summarize observed demand, satisfaction, admission rates, billing, and listed staff. Cluster numbers have no intrinsic order or meaning; the profile averages must be read to interpret them.

	Clusters	Silhouette score
1	3	0.411
0	2	0.400
3	5	0.312
2	4	0.302

patient_cluster	patients	mean_age	chronic_history_known_share	documented_chronic_condition_share	mean_appo
0	1117	46.236	0.000	0.000	
1	2920	45.039	1.000	1.000	
2	406	46.988	0.966	0.966	

facility_cluster	facilities	mean_appointments	mean_satisfaction	mean_admission_rate	mean_billed_amount
0	8	260.750	2.949	0.105	154783.172
1	5	224.600	2.860	0.120	160154.917
2	7	255.857	2.945	0.131	149628.156

department_cluster	departments	mean_appointments	mean_satisfaction	mean_admission_rate	mean_billed_amou
0	100	24.91	2.961	0.119	154504.81
1	50	24.56	2.850	0.125	152464.91
2	50	25.62	2.972	0.108	154263.20

Interpretation of cluster profiles. The patient silhouette score is highest for $k = 3$ (0.411), just above $k = 2$ (0.400); scores are lower for $k = 4$ (0.302) and $k = 5$ (0.312). The three groups are moderately separated in the standardized feature space. Their profiles show different data and service patterns:

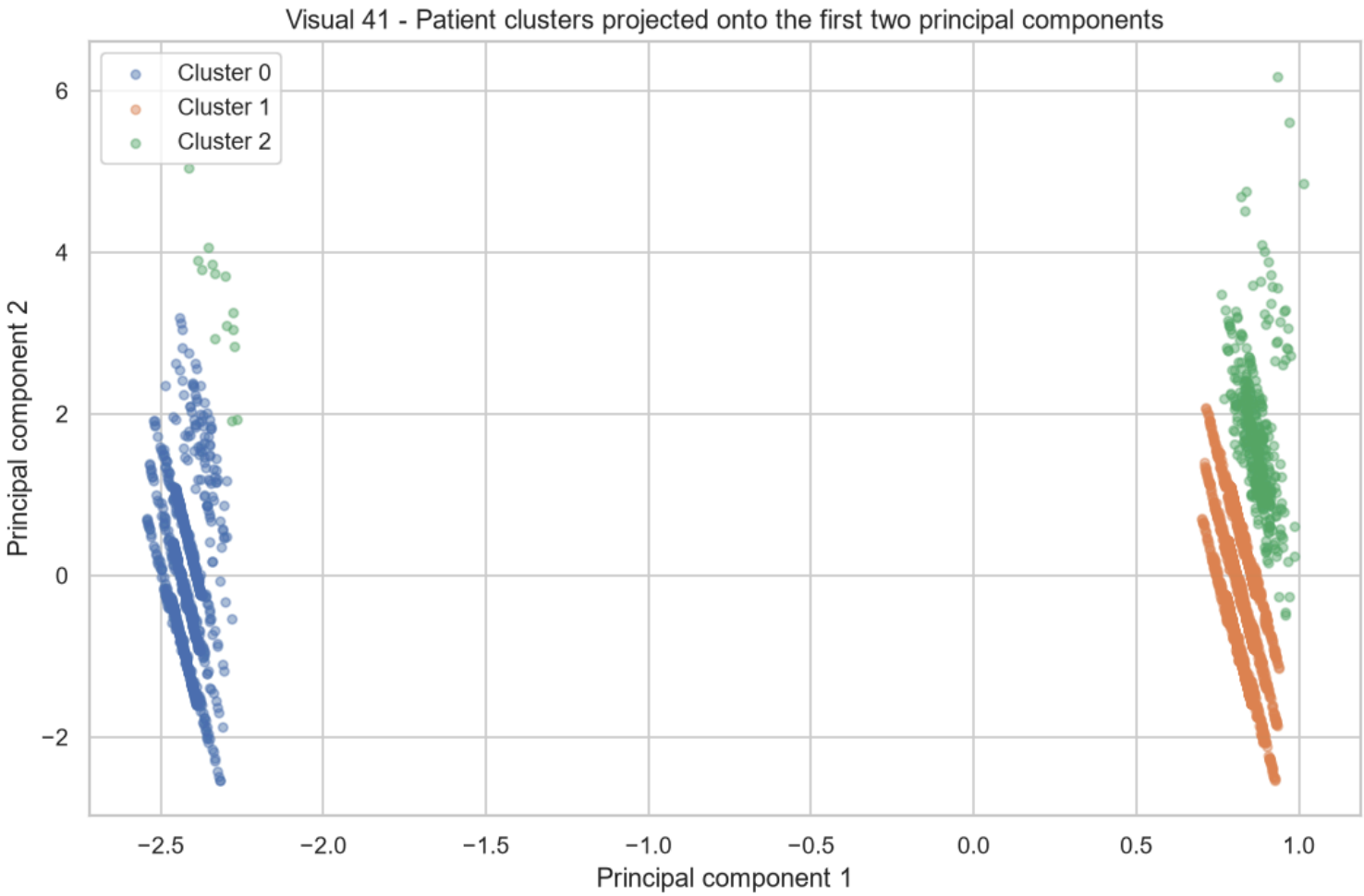
- **Cluster 0: 1,117 patients (25.1%).** No chronic history is recorded for this group. Only 30.4% have a classifiable high-risk proxy, and every classifiable case is flagged because age is 65+. The remaining histories are unknown; this cluster must not be labeled low-risk or universally high-risk
- **Cluster 1: 2,920 patients (65.7%).** Chronic history is documented for all patients in this group. They average one observed appointment and 0.116 admissions per patient. The screening proxy is positive, but it reflects documented chronic history and is not a validated clinical risk score
- **Cluster 2: 406 patients (9.1%).** This group averages 2.116 appointments and 0.276 admissions per patient; 96.6% have a documented chronic condition. The proxy is classifiable for 98.0% and positive for all those cases. This repeat-use profile may merit continuity-of-care review, but it does not prove clinical high risk or predict future admission.

The facility clusters contain 8, 5, and 7 facilities. Their mean satisfaction values are 2.949, 2.860, and 2.945; appointment volumes average about 261, 225, and 256; admission rates are 10.5%, 12.0%, and 13.1%. These are descriptive profiles, not facility rankings. The 94.9% DepartmentID/FacilityID mismatch may compromise facility assignment.

The department clusters contain 100, 50, and 50 departments. Average satisfaction is 2.961, 2.850, and 2.972, while average appointment volume is 24.91, 24.56, and 25.62. The middle group has three listed nurses on average, compared with four in the first group; actual shifts and workloads are unavailable. These patterns are discussion points for operations teams, not evidence that staffing caused rating differences. Facility and department cluster counts were set to three for compact profiling rather than selected by a validation score.

7.5 - Optional Principal Component Analysis (PCA)

PCA creates new, uncorrelated axes from weighted combinations of the standardized patient features; the first axis captures the greatest possible variance and the second captures the greatest remaining variance. The two-axis chart is an exploratory projection, not a replacement for the original variables or a clinical mechanism.




```
Variance explained by component 1: 33.4%
Variance explained by component 2: 17.3%
Variance explained by the two components: 50.7%
```

Visual 41. The PCA projection shows how the three exploratory patient clusters separate on the first two components.

Interpretation of PCA. The first component explains 33.4% of standardized feature variance and the second 17.3%, for 50.7% together. The two-dimensional plot therefore omits 49.3% of the variation in the original patient features. Some overlap is expected in this projection; it does not show the full cluster geometry. PCA is a visualization aid, not a prediction model or a clinical explanation.

7.6 - Predictive Modeling Summary

The admission classifier performs poorly on its holdout set (ROC AUC 0.466; balanced accuracy 0.468), and the satisfaction classifier is near chance (20.1% balanced accuracy across five ratings). Neither is ready for operational decisions. For overall monthly series, the three-month mean has lower one-step backtest errors than the fixed-parameter Holt damped-trend model for visits, gross bills, and receipts; however, the test uses only three months and does not establish 12-month accuracy. Patient clustering selects three groups by silhouette score. One group is dominated by missing chronic histories, another by documented chronic conditions, and a smaller group combines repeat visits with higher recorded admission frequency. None is a clinical risk label. Revisit the models with longer, temporally separated data and validated source relationships.

Part 8: Business Insights & Strategic Recommendations

This section translates the EDA, data validation, and model outputs into management actions. Each recommendation is tied to an observed finding and includes a practical measure for follow-up. Data-quality constraints are treated as part of the business problem because they affect the reliability of facility, financial, and patient-experience comparisons.

8.1 - Operational Challenges and Data Reliability

The first priority is to resolve source inconsistencies that limit staffing, facility, and financial decisions. The recommendations below distinguish directly observed evidence from items requiring operational verification.

Evidence-led Operational Priorities

Priority	Evidence in this analysis	Recommended action and follow-up measure
Reconcile location identifiers	4,744 of 5,000 appointments (94.9%) disagree on the facility mapped to their department.	Validate the DepartmentID-FacilityID master mapping. Track the share of appointments with a verified mapping before publishing facility comparisons.
Improve patient feedback and service recovery	Ratings are linked to 1,756 appointments (35.1%); 719 of these (40.9%) are 1 or 2, and the mean is 2.925/5.	Increase response coverage and route low ratings for review. Track response rate, low-rating share, and complaint-closure time.
Complete chronic histories	1,131 of 4,443 patient histories (25.5%) are Not recorded; 1,471 appointment rows involve patients age 65+ (29.4%). The engineered risk marker has no confirmed zero group.	Add a structured history review. Track completeness and follow-up for older patients; do not treat the proxy as a validated risk prevalence.
Reconcile bills and transactions	323 bills marked Paid still show a balance; 1,626 marked Pending have linked payments. The transaction-derived outstanding balance is 76,582,119 source units.	Reconcile invoice status with payment transactions. Track unreconciled bills, balance aging, and payment-allocation rate.
Plan staffing with better workload data	Department workload is only a roster proxy; its 10th and 90th percentiles are 10.5 and 89.175. Listed bed status does not measure occupancy.	Combine appointment demand with actual rosters, shifts, bed occupancy, and equipment use. Track forecast error and service coverage by shift.
Assess expansion with access evidence	There are 415 referral-linked appointments, but no travel-time or verified service-area measure in this extract.	Combine patient origins, referral completion, facility capacity, and travel times before choosing locations or services. Track population within target travel time and referral completion by district.
Improve financial reporting	Forecasts use gross bills and linked receipts, not reconciled accounting revenue; currency is unspecified and Revenue_Summary is unlinked.	Reconcile the finance sources before setting revenue targets. Track billed totals, receipts, and validated balances separately.

Interpretation. The evidence points first to data reliability: facility comparisons are affected by identifier mismatches, and financial balances are affected by status/transaction conflicts. Patient experience and record completeness are the next actionable areas. Forecasts and clusters can guide questions for managers, but current validation is too limited to justify staffing, expansion, or clinical decisions on their own.

8.2 - High-Risk Groups and Patient Experience

The patient profile contains 4,443 distinct people. Chronic conditions are documented for 3,312 (74.5%), while 1,131 histories (25.5%) are Not recorded. The 1,471 appointment rows in the age-65+ band represent 29.4% of the appointment cohort, not 29.4% of distinct patients. In the patient clustering, the 406-person repeat-use group averages 2.116 appointments and 0.276 recorded admissions per patient; this can inform continuity-of-care review,

but neither the cluster nor the engineered screening marker is a clinical diagnosis. The first priority is complete, dated health histories and clinician-led review.

Only 35.1% of appointments have linked feedback. Among the 1,756 ratings, 719 (40.9%) are 1 or 2 and the mean is 2.925/5. Increase feedback coverage and investigate low-scoring experiences. Because responses are selective and facility mapping is inconsistent, ratings should not be used as unadjusted facility or staff rankings.

8.3 - Resource Optimization and Revenue Improvement

The three-month mean baseline projects about 412 visits, 62.81 million in gross billed amounts, and 56.84 million in recorded receipts per month. In the three-month backtest it has lower MAE than Holt's damped-trend model for all three series, but this short validation cannot establish a reliable annual forecast. Department forecasts put DEP115 (Obstetrics & Gynecology) and DEP101 (Emergency) highest at 56 visits each over the next year; use these values to start a manager review, not to set staffing levels.

The department workload index uses listed staff counts and cannot represent shift coverage. The source bed table marks beds available but contains no occupancy events, so it cannot validate bed utilization. The 76.58 million calculated outstanding balance should be reconciled from transactions; the amount is in unspecified source units, and the unlinked Revenue_Summary cannot confirm recognized revenue. Track gross bills, paid transactions, and validated outstanding amounts separately.

8.4 - Strategic Recommendations and Management Follow-up

1. **Repair data lineage.** Assign owners and deadlines to validate department/facility mappings and reconcile billing statuses against transactions.
2. **Build a patient-experience feedback loop.** Increase response coverage, investigate low ratings, and document service recovery by department after location mappings are verified.
3. **Improve longitudinal patient records.** Collect structured chronic-condition history and dated updates; use the current screening proxy only to direct review, not to make clinical decisions.
4. **Use forecasts as rolling planning references.** Compare monthly actuals with both the moving-average baseline and Holt model. Reassess once several years of monthly data are available.
5. **Plan staffing from observed capacity.** Combine demand with shift rosters, bed occupancy, and equipment utilization before reallocating resources.
6. **Assess expansion using access evidence.** Combine geographic demand, referral completion, travel times, and facility capacity before selecting underserved districts or services.
7. **Govern model use responsibly.** Neither classification model is ready for operational decisions. Monitor performance across patient groups, validate with future-time data, and keep clinical decisions with qualified clinicians.

These actions focus on source reliability, patient experience, capacity planning, and financial transparency. Clinical, operations, finance, and data teams should review the proposed measures together each month.

Final Deliverable: Export Cleaned and Engineered Data

This final step saves the cleaned appointment-level master dataset and the engineered feature table for reuse outside the notebook. The master dataset is the quality-controlled integration output; engineered features retain documented proxy labels and unknown categories where source evidence is incomplete.

Export check: The two CSV files are stored in `Data/Cleaned_Datasets`. The first contains the cleaned 5,000-row appointment master; the second adds analysis-ready features while preserving explicit unknown and proxy

categories.